Complex Hamiltonian learning, KAM stability and safe magnetic actuation of a clamped–free catheter model

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Abstract

Stability guarantees for magnetic catheter robots require consistent mechanics, learned dynamics and actuation constraints. We develop a framework for a clamped–free rod model motivated by robotic intervention in microgravity. A variational formulation leads to a consistent-mass beam retaining the full magnetic potential. A complex Hamiltonian neural network learns conservative energy and nonnegative dissipation through a power-consistent decoder. Explicit statistical and deterministic certificates connect nonzero derivative loss to stability hypotheses. Rigorous a posteriori Kolmogorov–Arnold–Moser (KAM) computations establish invariant tori with reconstructed amplitudes near $1.9\,\mathrm{mm}$ for the target and five actual learned energies, including one identified from COMSOL displacement and velocity alone; nondegeneracy yields positive-measure families nearby. For the unforced physical finite-element model, strict convexity additionally establishes global asymptotic stability under positive damping. Independently constructed COMSOL and Python models agree at the matrix and trajectory levels. A continuum energy inequality bounds the entire centreline, and a scalar magnetic input governor enforces this bound. Under a $10\,\mathrm{mT}$ sinusoidal request, the uncontrolled trajectory exceeds $4\,\mathrm{mm}$ clearance, whereas the controlled trajectory remains within a certified $3.899\,\mathrm{mm}$ envelope. These results connect conservative invariant dynamics, dissipative stability and safe actuation for one explicitly declared mechanical model.

Keywords

magnetic catheter robots, complex Hamiltonian neural networks, KAM theory, finite elements, dissipation, safety control

1 Introduction

Magnetic actuation supplies remote forces and torques to small flexible robotic devices, including tools for minimally invasive intervention (Abbott et al. 2020; Kim et al. 2019). Their dynamics couple elasticity, magnetic fields, surrounding fluid and wall interaction. Long-duration space missions motivate investigating this interaction in reduced gravity. Documented spaceflight-associated fluid redistribution and neuro-ocular changes (NASA n.d.) provide application context; the present work studies mechanics and computation rather than a clinical or in-orbit procedure.

Hamiltonian neural networks (HNNs) learn dynamics through a scalar energy (Greydanus et al. 2019). Dissipative and port-Hamiltonian extensions add losses and external work (Zhong et al. 2020; Desai et al. 2021). Complex-valued networks provide coupled amplitude and phase representations (Trabelsi et al. 2018; Zhang et al. 2022), while KAM theory establishes persistence of quasiperiodic invariant motion under appropriately small Hamiltonian perturbations (Pöschel 2001; Bounemoura 2020). Combining these methods requires a quantitative bridge from the fitted model to the derivative norms, nonlinear twist and nonresonance hypotheses of a KAM theorem. A low trajectory error alone does not supply that bridge.

An equally important issue is mechanical consistency across the chain. A polynomial oscillator may permit a complete KAM calculation yet fail to represent the boundary conditions or magnetic actuation of the rod being studied. We therefore use one clamped–free magnetic beam for independent finite-element assembly, modal reduction, complex learning, nonlinear normal forms and dimensional safety calculations. The magnetic potential is retained as a full cosine. Its quartic expansion supplies twist; higher terms remain in the dynamics and in the analytic remainder. A constant bias field belongs to the conservative energy,

whereas changing the transverse field supplies work through a physical scalar port.

The contributions are fourfold. First, the distributed rod and its reduced complex decoder retain a consistent energy and power balance. Second, nonzero derivative loss is related to uniform Hamiltonian error through statistical estimates and an executable finite-feature certificate. Third, the nonlinear stability calculation concerns the actual magnetic model and fitted energy, with one fixed canonical chart. In addition to local normal forms, rigorous a posteriori certificates establish invariant tori at millimetre-scale reconstructed amplitudes. One saved learned function satisfies both a nonzero-loss twist certificate and the finite-amplitude certificate. Fourth, a continuum clearance theorem and a magnetic input governor connect dimensional actuation limits to noncollision conditions without assuming that a two-mode truncation bounds the full beam. Independent COMSOL computations test the mechanical implementation and expose the size of modal truncation error separately from learning error.

2 Distributed mechanics and the unified magnetic model

2.1 Rod mechanics and a variationally consistent reduction

Following the spatial rod formulation (Simo 1985; Antman 2005), let $s \in [0, L]$ denote reference arclength, $r(s,t) \in \mathbb{R}^3$ the centreline and $D(s,t) \in SO(3)$ its director frame. Define

$$\Gamma = D^{\mathsf{T}} r_s, \quad D_s = D\widehat{\kappa}, \quad D_t = D\widehat{\omega}_b. \tag{1}$$

The hat map is defined by $\widehat{a}b = a \times b$. The extensible, unshearable Kirchhoff specialization has $\Gamma = (0, 0, 1+e)^{\mathsf{T}}$; inextensibility additionally sets $e = 0$. Transverse internal forces are then constraint reactions. They must not be set to zero by substituting zero shear into a shearable constitutive law. In spatial components the balances are

$$\rho A r_{tt} = n_s + f_{\mathrm{mag}} + f_{\mathrm{h}} + f_{\mathrm{c}} + \rho A g, \tag{2}$$

$$\partial_t (D \mathcal{J}_b \omega_b) = m_s + r_s \times n + \tau_{\mathrm{ext}}. \tag{3}$$

For constant body inertia, the left side of the second equation is $D(\mathcal{J}_b \dot{\omega}_b + \omega_b \times \mathcal{J}_b \omega_b)$. With stored energy density $W(\Gamma, \kappa)$, unconstrained components of $D^{\mathsf{T}} n$ and $D^{\mathsf{T}} m$ are its strain derivatives. A fixed base specifies position and orientation; a free tip specifies the appropriate force and moment resultants, including any concentrated magnetic load.

For a body-fixed magnetic moment density $\mu_b(s)$ and a static applied field $B_0(r)$, the magnetic potential is

$$V_{\mathrm{mag}} = -\int_0^L (D\mu_b) \cdot B_0(r)\, ds. \tag{4}$$

Its translational and rotational variations give $f_{\mathrm{mag}} = \nabla_r (\mu \cdot B_0)$ at fixed orientation and $\tau_{\mathrm{mag}} = \mu \times B_0$, where $\mu = D\mu_b$. Thus static magnetic loading belongs to the conservative energy. Field modulation supplies explicit time-dependent work or an input port. A fixed-orientation dipole approximation is equivalent to a consistent Maxwell-stress calculation in its regime of validity; the same load is not added twice.

More explicitly, the magnetic force on a finite material segment can be evaluated on a surrounding vacuum or nonmagnetic control surface Σ using

$$F_{\mathrm{mag}} = \int_\Sigma T_M n_\Sigma \, \mathrm{d}A, \qquad T_M = \mu_0^{-1}(BB^{\mathsf{T}} - \tfrac{1}{2}|B|^2 I). \tag{5}$$

Here B is the total field on the control surface, including the self-consistent magnetic contribution of the material. The associated moment is the integral of the moment arm crossed with $T_M n_\Sigma$. The dipole formulas above are its small-magnet reduction when the applied field varies slowly across the magnetic segment (Abbott et al. 2020).

A modified Morison-type local closure, stated per unit reference length, is

$$f_{\mathrm{h}} = m_f a_f + m_a(a_f - r_{tt}) - c_1 v_r - c_2 |v_r| v_r, \tag{6}$$

$$v_r = r_t - v_f, \qquad c_1, c_2 \geq 0. \tag{7}$$

Here $m_f = \rho_f A_f$ and m_a are displaced-fluid and added mass per unit length; a_f is the prescribed fluid acceleration. This is a reduced closure requiring calibration, not a solved Navier–Stokes interaction. In still fluid its drag power is $-c_1|r_t|^2 - c_2|r_t|^3$. In moving fluid it is dissipative relative to the fluid velocity and can do positive work on the rod. A constant added mass may be moved to the inertial side exactly once.

For a stationary wall, define penetration $\delta(r)$ positive inside the wall and $n_c = \nabla\delta$. One nonadhesive penalty model is

$$V_c = \frac{k_c}{m+1}[\delta]_+^{m+1}, \tag{8}$$

$$f_c = -k_c[\delta]_+^m n_c - c_c \mathbf{1}_{\{\delta>0\}}[\dot{\delta}]_+ n_c. \tag{9}$$

The elastic term is included through V_c and the second term dissipates power during compression. Friction can be added as a dissipative port. This switching law is used for force

accounting outside the contact-free KAM domain, not as an analytic perturbation through an impact.

Choose a configuration map $(r(s,q), D(s,q))$ that satisfies the essential boundary conditions and the rod constraints. A linear centreline ansatz alone is insufficient to specify rotational inertia. Define

$$r_i = \partial_{q_i} r, \qquad a_i = \mathrm{vee}(D^\mathsf{T} \partial_{q_i} D). \tag{10}$$

Then $\omega_b = \sum_i a_i \dot{q}_i$ and the reduced kinetic energy is

$$T_N = \tfrac{1}{2} \dot{q}^\mathsf{T} M(q) \dot{q}, \tag{11}$$

$$M_{ij}(q) = \int_0^L \left(\rho A r_i \cdot r_j + a_i^\mathsf{T} \mathcal{J}_b a_j \right) ds. \tag{12}$$

An added-fluid-mass contribution may be included when its consistent kinetic metric and associated inertial terms are specified. Otherwise hydrodynamic inertia remains in the generalized force balance and is not also included in M. Galerkin projection of the virtual-work identity gives

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + \nabla V_N(q) = Q_{\mathrm{nc}}, \tag{13}$$

$$Q_i = \int_0^L (f \cdot r_i + \tau \cdot D a_i) ds + Q_i^\partial. \tag{14}$$

Here $C\dot{q}$ denotes the Christoffel contribution of M, and V_N contains elastic, static magnetic and retained gravitational potentials. Microgravity reduces the last contribution; it does not eliminate fluid drag or establish a particular blood-flow profile. Relative-flow drag, moving-base work and contact losses are retained in Q_{nc}. A stationary-fluid quadratic drag, for example, has nonpositive power; a moving fluid may inject energy. Contact switching is excluded from the analytic neighbourhood used for KAM analysis.

For $M(q) \succ 0$, the Legendre transform yields

$$p = M(q)\dot{q}, \qquad H_N(q,p) = \tfrac{1}{2} p^\mathsf{T} M(q)^{-1} p + V_N(q). \tag{15}$$

In particular, $\dot{p}_i = -\tfrac{1}{2} p^\mathsf{T} \partial_{q_i}(M^{-1}) p - \partial_{q_i} V_N + (Q_{\mathrm{nc}})_i$. The configuration dependence of inertia is retained. Potential forces appear either in V_N or in Q_{nc}, never both.

2.2 Complex energy–dissipation representation with an explicit decoder

Energy and Rayleigh dissipation have different physical units. Let E_0 and t_0 be reference energy and time, and choose diagonal position and momentum scales Q_0, P_0 with $Q_0^\mathsf{T} P_0 = E_0 t_0 I$. Set $q = Q_0 x$, $p = P_0 y$, $\tau = t/t_0$ and

$$h(x,y) = H_N(Q_0 x, P_0 y)/E_0, \tag{16}$$

$$d(x,v) = \frac{t_0}{E_0} \mathcal{D}_N(Q_0 x, Q_0 v/t_0). \tag{17}$$

Here $v = dx/d\tau$ and $\mathcal{D}_N$ is a Rayleigh dissipation potential. The two entries of $h + id$ are now dimensionless. A complex network provides coupled features and two physically decoded real readouts,

$$\mathcal{H}_\theta(x,y,v) = h_\theta(x,y) + i d_\theta(x,v), \tag{18}$$

$$h_\theta = h_{\mathrm{prior}} + h_{\mathrm{res},\theta}. \tag{19}$$

One admissible dissipative head is

$$d_\theta(x,v) = \tfrac{1}{2} v^\mathsf{T} D_\theta(x) v, \qquad D_\theta(x) = L_\theta(x) L_\theta(x)^\mathsf{T} \succeq 0. \tag{20}$$

More general heads may be used if $d_\theta(x,\cdot)$ is differentiable, convex and minimized at zero. The decoder is

$$v = \partial_y h_\theta, \qquad x' = v, \tag{21}$$

$$y' = -\partial_x h_\theta - \partial_v d_\theta(x,v) + B(x)u. \tag{22}$$

The derivative of d_θ is taken with respect to its independent velocity argument before substituting $v = \partial_y h_\theta$. It is not a total derivative through this substitution.

Proposition 1. Power-consistent complex representation. *For a differentiable h_θ, the decoded system satisfies*

$$\frac{dh_\theta}{d\tau} = -v^\mathsf{T} \partial_v d_\theta(x,v) + u^\mathsf{T} B(x)^\mathsf{T} v + \partial_\tau h_\theta. \tag{23}$$

For the quadratic head, the first term equals $-v^\mathsf{T} D_\theta(x) v \le 0$. For the convex head it is also nonpositive.

Proof. Apply the chain rule to h_θ and substitute (22). The two canonical terms cancel. Convexity and the minimum at zero imply $v^\mathsf{T} \partial_v d_\theta(x,v) \ge d_\theta(x,v) - d_\theta(x,0) \ge 0$.

With $z = (x + iy)/\sqrt{2}$, the canonical convention is $\{z_j, \bar{z}_k\} = -i\delta_{jk}$ and

$$z' = -i\partial_{\bar{z}} h_\theta - \frac{i}{\sqrt{2}} \partial_v d_\theta + \frac{i}{\sqrt{2}} B(x) u. \tag{24}$$

This is a unified complex representation of conservative and nonconservative dynamics with a specified physical decoder. The rule $z' = -i\partial_{\bar{z}}(h_\theta + i d_\theta)$ would generally produce a different vector field and has no automatic dissipativity guarantee. Real analyticity in (x,y), and a controlled analytic extension to a complex neighbourhood, are the relevant requirements for analytic KAM. Holomorphicity

Table 1. Declared parameters of the unified magnetic cantilever. These are illustrative mechanical parameters, not measurements from a clinical or spaceflight device.

Quantity	Symbol	Value
Length	L	$0.08\,\mathrm{m}$
Circular radius	r_c	$0.0006\,\mathrm{m}$
Young's modulus	E	$5.0 \times 10^6\,\mathrm{Pa}$
Density	ρ	$1100\,\mathrm{kg\,m^{-3}}$
Tip dipole	μ	$2.0 \times 10^{-5}\,\mathrm{A\,m^2}$
Axial bias field	B_0	$0.03\,\mathrm{T}$
Mass damping rate	α	$0.55558524\,\mathrm{s^{-1}}$
Transverse field amplitude	B_a	$0.006\,\mathrm{T}$
Drive frequency	Ω_d	$9.4449490\,\mathrm{rad\,s^{-1}}$
Ramp time	t_r	$0.25\,\mathrm{s}$
Uniform Hermite elements	N_e	32

in z alone is not required of a nonconstant real energy $h_\theta(z, \bar{z})$. In particular, ReLU or modReLU features require a separate smooth approximation or a theorem of appropriate regularity; their ordinary piecewise-smooth implementation is not analytic.

The dimensional specialization below uses explicit SI units. The learning and KAM statements use dimensionless canonical variables, with their energy and time scales declared before fitting.

2.3 One clamped–free magnetic beam throughout the computational study

We instantiate the rod reduction as a straight, planar, small-slope Euler–Bernoulli cantilever with a concentrated magnetic dipole at its free tip. The base position and slope are fixed. The elastic part is linear in curvature; the full trigonometric magnetic potential is retained. These choices define a reproducible mechanical specialization of the spatial rod, rather than a catheter-specific material identification. Table 1 states all dimensional parameters. Gravity, imposed fluid flow and field gradient are zero in this specialization. Still-fluid/structural losses are represented by a mass-proportional Rayleigh term. The more general force accounting above remains applicable when those terms are present.

Write $w(s,t)$ for transverse displacement and $\vartheta(t) = w_s(L,t)$ for the small-slope tip rotation. A uniform field $(B_0, B_y(t))$ has zero translational dipole force and the tip moment

$$M_{\mathrm{tip}} = \mu\{B_y(t)\cos\vartheta - B_0\sin\vartheta\}. \tag{25}$$

This is the natural moment boundary load. It does not impose a second kinematic constraint at the free tip. The magnetic potential is $-\mu(B_0\cos\vartheta + B_y\sin\vartheta)$, so that (25) follows by variation. Keeping this potential exact does not remove the small-slope approximation in the beam kinematics.

Let $q = (w_1, \vartheta_1, \ldots, w_{N_e}, \vartheta_{N_e})^{\mathsf{T}}$ contain the unconstrained cubic-Hermite nodal coordinates, e_ϑ select its final rotation, and M, K be the consistent mass and bending stiffness matrices. The physical equations and static-bias energy are

$$M\ddot{q} + \alpha M\dot{q} + Kq = e_\vartheta M_{\mathrm{tip}}, \tag{26}$$

$$H_0(q,p) = \tfrac{1}{2}p^{\mathsf{T}}M^{-1}p + \tfrac{1}{2}q^{\mathsf{T}}Kq + E_0\{1-\cos(e_\vartheta^{\mathsf{T}}q)\}, \tag{27}$$

$$E_0 = \mu B_0 = 6\times10^{-7}\,\mathrm{J}, \qquad p = M\dot{q}. \tag{28}$$

There is no axial end restraint and no artificial stretching quartic. The nonlinearity used in learning and in the KAM calculation is the magnetic nonlinearity of this same cantilever.

The static-field linearization is $K_b = K + E_0 e_\vartheta e_\vartheta^{\mathsf{T}}$. Choose $\Phi^{\mathsf{T}}M\Phi = I$ and $\Phi^{\mathsf{T}}K_b\Phi = \mathrm{diag}(\omega_j^2)$, with positive tip-slope entries fixing modal signs. For n retained modes, $q = \Phi_n a$, and the modal momentum is $\pi = \dot{a}$. Set $\omega_0 = \omega_1$, $\tau = \omega_0 t$, $\nu_j = \omega_j/\omega_0$, and

$$a_j = \sqrt{\frac{E_0}{\omega_0\omega_j}}X_j, \qquad \pi_j = \sqrt{\frac{E_0\omega_j}{\omega_0}}Y_j. \tag{29}$$

The product of each position–momentum scale is E_0/ω_0; division of energy by E_0 and rescaling time therefore give the standard dimensionless canonical equations. Define

$$S = \Phi_n\,\mathrm{diag}\sqrt{\frac{E_0}{\omega_0\omega_j}}, \quad d = S^{\mathsf{T}}e_\vartheta, \quad \ell(s)^{\mathsf{T}} = N(s)S, \tag{30}$$

where $N(s)$ is the Hermite displacement row. Thus the full centreline and tip slope are recovered in SI units as $w(s) = \ell(s)^{\mathsf{T}}X$ and $\vartheta = d^{\mathsf{T}}X$. The single dimensionless Hamiltonian used below is

$$h_*(X,Y) = h_2(X,Y) + R_{\mathrm{mag}}(X), \tag{31}$$

$$h_2 = \tfrac{1}{2}\sum_{j=1}^{n}\nu_j(X_j^2+Y_j^2), \tag{32}$$

$$R_{\mathrm{mag}} = 1 - \cos(d^{\mathsf{T}}X) - \tfrac{1}{2}(d^{\mathsf{T}}X)^2. \tag{33}$$

The subtracted quadratic avoids counting the static magnetic stiffness twice. For the two-mode model, $(\omega_1,\omega_2) = (11.41728728, 70.24049294)\,\mathrm{rad\,s^{-1}}$. All learned models use this fixed chart; states are not independently renormalized after fitting.

With $u = B_y/B_0$, $\gamma = \alpha/\omega_0$ and $\mathsf{N} = \mathrm{diag}(\nu_j)$, the physical port decoder becomes

$$X' = \partial_Y h, \tag{34}$$
$$Y' = -\partial_X h - \gamma \mathsf{N}^{-1}\partial_Y h + u\, d\cos(d^{\mathsf{T}}X). \tag{35}$$

For $h = h_*$, the damping term is exactly $-\gamma Y$. The Rayleigh head is $\mathcal{D}(v) = \frac{1}{2}\gamma v^{\mathsf{T}}\mathsf{N}^{-1}v$. For a learned energy, retaining the velocity-based form in (35) preserves nonnegative dissipation. Using $-\gamma Y$ after learning a momentum-dependent residual would, in general, lose that guarantee. The corresponding power identity is

$$h' = -\gamma(\partial_Y h)^{\mathsf{T}}\mathsf{N}^{-1}\partial_Y h + u\cos(d^{\mathsf{T}}X)d^{\mathsf{T}}\partial_Y h. \tag{36}$$

Alternatively, including $-u\sin(d^{\mathsf{T}}X)$ in the time-dependent energy moves the input power to $-u'\sin(d^{\mathsf{T}}X)$. These are equivalent bookkeeping conventions.

2.4 Independent COMSOL realization of the same discretization

The model was also constructed and solved in COMSOL Multiphysics 6.3.0.290 through LiveLink for MATLAB. Its two-dimensional Beam interface uses the Euler–Bernoulli formulation, 32 uniform elements, a consistent mass matrix, a fixed base, and the nonlinear tip moment (25). Geometric nonlinearity is disabled to match the declared small-slope mechanical model. The zero axial displacement subspace decouples, leaving 64 transverse displacement/rotation degrees of freedom.

After selecting transverse degrees of freedom by their field names and coordinates, the independently assembled COMSOL and Python mass and elastic stiffness matrices have relative Frobenius differences 7.76×10^{-15} and 3.05×10^{-14}, respectively. The first six modal frequencies agree to a maximum relative difference of 1.88×10^{-9}. These checks establish agreement of the stated discretizations; they do not establish the material validity of the chosen parameters. The native model, assembly exports, parameter values and time-domain verification records accompany the manuscript.

2.5 A global result for the unforced physical model

Proposition 2. Unforced global stability of the full FE target. *For the full clamped–free Hermite model (27), set $B_y = 0$. With the declared parameters, the potential is globally strongly convex and the origin is its unique equilibrium. If $\alpha > 0$, the origin is globally asymptotically stable. If $\alpha = 0$, every finite-energy trajectory remains in a bounded energy sublevel and the origin is Lyapunov stable. These are global statements for the declared small-slope mathematical model; they do not establish its physical validity at large deflections or a global property of arbitrary learned quartic residuals.*

Proof. Write $b = e_\vartheta$ and $U(q) = q^{\mathsf{T}}Kq/2 + E_0\{1 - \cos(b^{\mathsf{T}}q)\}$. The clamped conditions give $|w_s(L)|^2 \le L\int_0^L |w_{ss}|^2\, ds$, including for every Hermite finite-element function. Equivalently, $bb^{\mathsf{T}} \preceq [L/(EI)]K$; equality of the corresponding tip compliance follows from the static quadratic bending shape. Thus, for every q,

$$D^2U(q) = K + E_0\cos(b^{\mathsf{T}}q)bb^{\mathsf{T}}$$
$$\succeq (1-\kappa)K \succeq 0.905K, \tag{37}$$
$$\kappa = \frac{E_0 L}{EI} = \frac{8}{27\pi} < 0.095. \tag{38}$$

In particular, $\nabla U = 0$ only at $q = 0$. Both M and K are positive definite, and

$$H_0 \ge \tfrac{1}{2}p^{\mathsf{T}}M^{-1}p + \tfrac{1}{2}q^{\mathsf{T}}Kq.$$

Its sublevels are compact. Along the unforced dynamics,

$$\dot{H}_0 = -\alpha p^{\mathsf{T}}M^{-1}p \le 0.$$

This proves boundedness, global continuation and Lyapunov stability. For $\alpha > 0$, the largest invariant set with $\dot{H}_0 = 0$ has $p = 0$ and $\nabla U(q) = 0$, hence consists only of the origin. LaSalle's invariance principle on each compact sublevel gives global asymptotic stability.

3 Complex Hamiltonian learning

3.1 A fixed canonical chart

For a positive definite constant mass M and reference stiffness K_{ref}, select U such that $U^{\mathsf{T}}MU = I$ and $U^{\mathsf{T}}K_{\mathrm{ref}}U = \Omega^2$, where $\Omega = \mathrm{diag}(\omega_1, \ldots, \omega_n)$. In the magnetic specialization, $K_{\mathrm{ref}} = K_b$ includes the static-field stiffness. The transformation

$$q = U\Omega^{-1/2}Q, \qquad p = MU\Omega^{1/2}P \tag{39}$$

is symplectic: its configuration and momentum blocks satisfy $S_q^{\mathsf{T}}S_p = I$. Consequently $H_2 = \frac{1}{2}\sum_j \omega_j(Q_j^2 + P_j^2)$. The computational variables in (29) obey $(Q, P) = \sqrt{E_0/\omega_0}(X, Y)$, together with the stated energy and time normalization. That fixed dimensionless chart is used for training, derivative errors and normal forms. Arbitrary independent standardization of q and p is not silently treated

as canonical; optional training weights change the loss metric only.

Write $x = (Q, P)$ and use one complex normalization throughout:

$$z_j = \frac{Q_j + iP_j}{\sqrt{2}}, \quad I_j = z_j \bar{z}_j, \qquad \{z_j, \bar{z}_k\} = -i\delta_{jk}. \quad (40)$$

Wirtinger calculus gives $\partial_{\bar{z}_j} = (\partial_{Q_j} + i\partial_{P_j})/\sqrt{2}$ and therefore $\dot{z} = -i\partial_{\bar{z}} H$ for real H. This is a change of representation of a real symplectic system. Holomorphicity in z alone would exclude most real energy functions, including $H_2 = \sum_j \omega_j z_j \bar{z}_j$. The analyticity required by analytic KAM instead means a holomorphic extension of the real variables (Q, P) to a complex neighbourhood, in which z and $\bar{z}$ are independent linear combinations.

3.2 Architecture and objective

A shared complex feature map $F_\theta(z, \bar{z}, b)$, with magnetic configuration b, drives real readouts for H_θ and a dissipation factor L_θ. The latter defines $\mathcal{D}_\theta(q, v) = \frac{1}{2} \left\| L_\theta(q)^\mathsf{T} v \right\|^2$. The physical decoder in Section 2 determines the vector field. An affine complex layer has the real block realization

$$(A + iB)(a + ib) = (Aa - Bb) + i(Ba + Ab). \quad (41)$$

Thus complex weights impose a specific coupling, rather than furnishing an extra physical state. Smooth split activations $\sigma(a + ib) = \tanh(a) + i \tanh(b)$ are real analytic on bounded real domains and extend to a sufficiently small pole-free complex neighbourhood when weights are bounded. Their use does not justify assuming an arbitrarily wide analytic strip.

For observed dimensionless states and velocities, a suitable objective is

$$\widehat{\mathcal{L}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \widehat{\dot{x}}_i - f_\theta(x_i, u_i) \right\|^p + \lambda_E \widehat{\mathcal{L}}_E + \lambda_R \mathcal{R}(\theta). \quad (42)$$

Energy observations, if available, fix a gauge through $\widehat{\mathcal{L}}_E$; vector-field training alone identifies H only up to a constant. To infer conservative gradient error from the full decoder loss, errors in dissipation and inputs must first be bounded or measured and included in the label-error budget. They cannot be absorbed into a Hamiltonian residual merely because their magnitudes are small.

The reproducible experiments specialize this architecture to a finite analytic complex feature network. The exact magnetic feature R_{mag} in (33) is represented by complex exponentials and a quadratic correction, with a learned real readout. Quartic features complement its degree-four Taylor coefficient; this keeps the full cosine target in the hypothesis class while avoiding a duplicate quartic direction. Products of complex affine forms and their conjugates generate all real quadratic and quartic monomials. They can be implemented by a finite network with polynomial activation, using $uv = ((u + v)^2 - (u - v)^2)/4$. A real readout combines the full magnetic analytic feature with the polynomial residual; conjugate features enforce the reality of the energy. The prescribed features exclude a spurious cubic term and, for the gray-box model, preserve the known static-field quadratic part exactly. Fitting the output coefficients from energy derivatives is linear least squares. The experiment therefore tests an explicitly specified finite-feature CHNN. The magnetic coefficient and residual readouts are fitted from data; no claim about optimization of an unrestricted deep network is needed.

3.3 What complex approximation theory establishes

Theorem 1 in the complex-reaction literature (Zhang et al. 2022) gives universal approximation for a specified reaction activation and an L^2 distributional criterion. Its comparison with real networks is tied to the stated function classes and resources. Such a theorem supplies neither higher derivative accuracy nor a universal advantage for magnetic catheter energies. In particular, piecewise reaction activations cannot be inserted unchanged into a smooth KAM proof.

There is nevertheless a direct smooth approximation route. A split-complex network with real-part output contains real smooth networks: for an input $(Q + iP)/\sqrt{2}$, choose first-layer complex weights $\sqrt{2}(A_Q - iA_P)$ to obtain real preactivation $A_Q Q + A_P P + b$. Subsequent real weights keep the real channel independent of the imaginary channel. Finite-order derivative approximation for smooth real networks (Hornik 1991) is thereby inherited as an existence result. Alternatively, on a compact box where a target has a convergent analytic power series, truncation gives a polynomial approximation with uniformly controlled derivatives. Those polynomials are realizable by the complex polynomial features used here. Training and a quantitative certificate are still required to establish that a particular fitted network attains the error needed for KAM.

4 From nonzero training loss to KAM persistence

4.1 A statistical bridge in the norm required by KAM theory

All Hamiltonians below are real-valued scalar functions, including when their parameters and hidden representations are complex. The realification of a complex parameter vector has dimension P. Let $\mathcal{D} = \mathbb{T}^n \times B$, where B is a bounded ball in action space, and write $d = 2n$. Lebesgue measure on $\mathcal{D}$ is denoted by dx. The following result completes the derivative-norm interface needed to combine the learning argument of Chen et al. (2022) with a finite-differentiability KAM theorem. It does not infer derivative smallness from function values alone.

Theorem 3. Uniform learning-to-C^ℓ bound. *Fix $d < p < \infty$, an integer $\ell \geq 1$, and an integer $s > \ell + d/p$. Let $X_1, \dots, X_N$ be independent samples with density $f_\mu \geq m_0 > 0$ on $\mathcal{D}$. Let $H_\star$ be the target Hamiltonian and $\{H_\vartheta : \vartheta \in \Theta\}$ the hypothesis class, with $\Theta \subset [-B_\Theta, B_\Theta]^P$. Assume that for all $x \in \mathcal{D}$ and $\vartheta, \eta \in \Theta$,*

$$\|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p \leq Q, \tag{43}$$

$$\|\nabla H_\vartheta(x) - \nabla H_\eta(x)\|_p \leq L_g \|\vartheta - \eta\|_\infty. \tag{44}$$

After fixing the additive gauge by

$$e_\vartheta = H_\vartheta - H_\star - \frac{1}{|\mathcal{D}|}\int_{\mathcal{D}} (H_\vartheta - H_\star)\, dx,$$

$$\sup_{\vartheta \in \Theta} \|e_\vartheta\|_{W^{s,p}(\mathcal{D})} \leq M_s, \tag{45}$$

suppose that M_s is a valid uniform bound. Define

$$\widehat{R}_N(\vartheta) = \frac{1}{N}\sum_{i=1}^{N} \|\nabla H_\vartheta(X_i) - \nabla H_\star(X_i)\|_p^p, \tag{46}$$

$$L_{\mathrm{par}} = pQ^{p-1}L_g, \qquad L_{\max} = Q^p. \tag{47}$$

For any fixed $0 < a \leq 2B_\Theta$ and $0 < \delta < 1$, put

$$\Gamma_N(a,\delta) = 2L_{\mathrm{par}}a + \frac{L_{\max}}{\sqrt{2N}}\sqrt{P\log(1 + 2B_\Theta/a) + \log(1/\delta)}, \tag{48}$$

$$S_N(\vartheta) = \widehat{R}_N(\vartheta) + \Gamma_N(a,\delta), \tag{49}$$

$$G_N(\vartheta) = C_{\mathrm{P}}\, m_0^{-1/p} S_N(\vartheta)^{1/p}, \tag{50}$$

$$\alpha = \frac{\ell - 1 + d/p}{s - 1} \in (0,1). \tag{51}$$

Here C_{P} is a Poincare constant chosen so that a mean-zero function satisfies $\|e\|_{W^{1,p}} \leq C_{\mathrm{P}}\|\nabla e\|_{L^p}$. With probability at least $1 - \delta$, simultaneously for every $\vartheta \in \Theta$,

$$\|e_\vartheta\|_{C^\ell(\overline{\mathcal{D}})} \leq E_N(\vartheta) := C_{\mathrm{GN}}\left[G_N(\vartheta) + M_s^\alpha G_N(\vartheta)^{1-\alpha}\right]. \tag{52}$$

C_{GN} depends only on the domain and (d, p, s, ℓ). We use $\|e\|_{C^\ell} = \max_{|a|\leq\ell} \sup |D^a e|$; equivalent conventions only change the displayed constants. The event is uniform over the hypothesis class and therefore includes parameters selected using these same training data.

Proof. The loss $r_\vartheta(x) = \|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p^p$ lies in $[0, L_{\max}]$ and is L_{par}-Lipschitz in the ℓ^∞ parameter metric. Partitioning the parameter cube into cells of side length a and choosing a representative from each nonempty cell gives an a-net with at most $(1 + 2B_\Theta/a)^P$ points. The one-sided Hoeffding inequality and a union bound control $R(\eta) - \widehat{R}_N(\eta)$ at every net point by the square-root term in Γ_N, where $R(\eta) = \int r_\eta\, d\mu$. The population and empirical losses each change by at most $L_{\mathrm{par}}a$ between a parameter and its net representative. Consequently $R(\vartheta) \leq S_N(\vartheta)$ uniformly.

The density lower bound gives $\|\nabla e_\vartheta\|_{L^p(dx)} \leq m_0^{-1/p} S_N(\vartheta)^{1/p}$. The gauge in (45) and Poincare's inequality then yield $\|e_\vartheta\|_{W^{1,p}} \leq G_N(\vartheta)$. Apply the Gagliardo–Nirenberg inequality to $g = \nabla e_\vartheta$, with derivative orders $j = \ell - 1$, $m = s - 1$, and both endpoint Lebesgue exponents equal to p. Its interpolation identity is

$$0 = \frac{j}{d} + \alpha\left(\frac{1}{p} - \frac{m}{d}\right) + \frac{1-\alpha}{p},$$

which gives the stated α. The strict inequality $s > \ell + d/p$ ensures $\alpha < 1$ and avoids the critical Sobolev endpoint. Extension from the smooth bounded action domain, together with periodic charts in the angular variables, gives

$$\|e\|_{C^\ell} \leq C_{\mathrm{GN}}\left(\|e\|_{W^{1,p}} + \|e\|_{W^{1,p}}^{1-\alpha}\|e\|_{W^{s,p}}^{\alpha}\right).$$

Lower-order derivatives are included by the inhomogeneous term and the $W^{1,p} \hookrightarrow C^0$ embedding, which uses $p > d$. Substitution proves (52).

For instance, one may select an integer $\ell > 2(\tau + 1)$, $p > 2n$, and $s = \ell + 1$. This makes every exponent explicit, although the resulting interpolation bound can be conservative. Bounds on M_s, Q, L_g follow from bounded parameters and derivatives of smooth activations on a compact domain; the corresponding bounds for $H_\star$ must also be justified. Merely declaring the functions smooth does not provide these quantitative bounds. Alternatively, a loss containing all spatial derivatives of the vector field through order ℓ controls $W^{\ell+1,p}$ directly, and Sobolev

embedding then gives a C^ℓ error with exponent $1/p$, without interpolation.

Corollary 4. KAM transfer for the target and the trained Hamiltonian. *Let $n \geq 2$, $\tau > n-1$ and an integer $\ell > 2(\tau+1)$ be fixed. Suppose that, in one specified canonical action–angle chart,*

$$H_\star = h_0(I) + P_\star(I,\varphi), \qquad H_\vartheta = h_0(I) + P_\vartheta(I,\varphi).$$

Assume $h_0 \in C^{\ell+2}(\overline{B})$ with finite norm, and $\nabla h_0 : B \to \Omega$ is a diffeomorphism with bounded inverse in C^ℓ. Fix $\gamma > 0$ and set

$$\Omega_{\gamma,\tau} = \left\{ \omega \in \Omega : \begin{array}{l} |k\cdot\omega| \geq \gamma |k|_1^{-\tau} \ (0 \neq k \in \mathbb{Z}^n), \\ \operatorname{dist}(\omega, \partial\Omega) \geq \gamma \end{array} \right\}. \tag{53}$$

Let $c_{\mathrm{K}} > 0$ be the constant in Theorem A of Bounemoura (2020) for this domain and integrable Hamiltonian. If a certified bound $\beta_\vartheta \geq \|P_\vartheta - \overline{P}_\vartheta\|_{C^\ell}$ satisfies

$$\beta_\vartheta + E_N(\vartheta) < c_{\mathrm{K}} \gamma^2, \tag{54}$$

then, on the event in Theorem 3, both $H_\star$ and H_ϑ possess invariant Lagrangian tori for every frequency in $\Omega_{\gamma,\tau}$. The flow on each torus is C^1-conjugate to the linear flow with that frequency. For a piecewise smooth frequency-domain boundary, the complement of each family has measure at most $C\gamma$. In particular, each family has positive measure when γ is small enough.

Proof. Write $\overline{P}$ for the spatial mean over $\mathcal{D}$. The gauge-fixed difference satisfies

$$P_\star - \overline{P}_\star = P_\vartheta - \overline{P}_\vartheta - e_\vartheta.$$

Thus the two perturbation norms are at most β_ϑ and $\beta_\vartheta + E_N(\vartheta)$, respectively. Constants added to a Hamiltonian do not change its vector field. Apply the cited finite-smoothness KAM theorem separately to the two Hamiltonians with the same h_0, domain, and Diophantine set. The theorem supplies tori of class $C^{\tau+1}$ with Lipschitz dependence as invariant graphs on frequency. The two systems need not have identical invariant embeddings.

The certified budget includes the physical perturbation from h_0, any normal-form remainder, and the learning error. A small learning error alone does not establish that the physical system is close to an integrable one. Theorem A in the published 2020 version of Bounemoura (2020) gives the measure bound $C\gamma$; the $C\gamma^2$ measure bound printed in the 2018 preprint should not be used.

4.2 A coefficient certificate for a finite analytic CHNN

The finite analytic readout used in the experiments yields a stronger certificate that avoids a dimension-dependent interpolation loss. Let $\psi_1, \ldots, \psi_m$ be real-valued basis functions on the real phase domain. Complex features may be converted into this basis by taking their real and imaginary parts and collecting equal monomials. The certificate applies to the effective coefficients of the actual trained Hamiltonian, irrespective of how they are parameterized inside the network.

Proposition 5. Deterministic global certificate from sampled gradients. *Suppose*

$$H_\vartheta = H_{\mathrm{prior}} + \sum_{j=1}^m c_{\vartheta,j}\psi_j,$$
$$H_\star = H_{\mathrm{prior}} + \sum_{j=1}^m c_{\star,j}\psi_j + R.$$

The basis contains no constant gauge direction. Let

$$G_N = \frac{1}{N}\sum_{i=1}^N D\psi(X_i)^\top D\psi(X_i),$$
$$\lambda_N = \lambda_{\min}(G_N) > 0,$$

where the jth column of $D\psi$ is $\nabla\psi_j$. For gradient observations $Y_i = \nabla H_\star(X_i) + \xi_i$, define

$$r_N = \left(\frac{1}{N}\sum_i \|\nabla H_\vartheta(X_i) - Y_i\|_2^2\right)^{1/2},$$
$$t_N = \left(\frac{1}{N}\sum_i \|\nabla R(X_i) + \xi_i\|_2^2\right)^{1/2}.$$

Then

$$\|c_\vartheta - c_\star\|_2 \leq \frac{r_N + t_N}{\sqrt{\lambda_N}}. \tag{55}$$

For any function norm $\|\cdot\|_{\mathcal{X}}$ satisfying the triangle inequality, put $K_{\mathcal{X}} = (\sum_j \|\psi_j\|_{\mathcal{X}}^2)^{1/2}$. If $\|R\|_{\mathcal{X}} \leq \varepsilon_R$, then

$$\|H_\vartheta - H_\star\|_{\mathcal{X}} \leq K_{\mathcal{X}} \frac{r_N + t_N}{\sqrt{\lambda_N}} + \varepsilon_R. \tag{56}$$

The norm may be C^ℓ on a real compact domain or the supremum norm on a prescribed complex neighborhood where every basis function and R is holomorphic. No independent sampling assumption is needed.

Proof. Let $\Delta c = c_\vartheta - c_\star$. The empirical norm of $D\psi\Delta c$ is $(\Delta c^\top G_N \Delta c)^{1/2}$. Since $D\psi\Delta c = (\nabla H_\vartheta - Y) + (\nabla R + \xi)$ at every sample, the triangle inequality gives $\sqrt{\lambda_N}\|\Delta c\|_2 \leq$

$r_N + t_N$. Finally, $\|\sum_j \Delta c_j \psi_j - R\|_{\mathcal{X}} \le K_{\mathcal{X}} \|\Delta c\|_2 + \varepsilon_R$ by Cauchy–Schwarz.

For raw monomials $\psi_\nu(x) = x^\nu$, an explicit C^ℓ bound on $[-r, r]^d$ is obtained from

$$\|\psi_\nu\|_{C^\ell} \le \max_{|a|\le \ell,\, a \le \nu} \frac{\nu!}{(\nu - a)!}\, r^{|\nu|-|a|}.$$

The analytic sup norm on $\{|x_j| \le r\} \subset \mathbb{C}^d$ is $r^{|\nu|}$. For all 35 degree-four monomials in four variables, $K_{\mathrm{an}} \le \sqrt{35}\, r^4$. If a canonical chart $x = \Phi(I, \varphi)$ maps an action–angle complex domain into this polydisc, the same analytic bound holds after composition. For a C^ℓ certificate after composition, the derivative bounds of Φ must instead be included by the chain rule. Unmodeled physical terms require the tail bounds t_N and ε_R. Their approximation-tail contribution is zero only when the target lies exactly in the declared class. The full-cosine feature class below has this property for its specified reduced target; it does not remove truncation error relative to the full beam.

The statistical result requires independent state samples with the stated coverage density. Consecutive states on a trajectory do not satisfy this assumption automatically. Accordingly, the trajectory-based experiments below use the deterministic Gram certificate, which requires no sampling independence. Their implementation divides both squared residuals and the Gram matrix by $4N$ rather than N; the coefficient bound is unchanged provided the noise norm uses that same scalar-observation normalization.

5 Nonlinear stability of the learned magnetic model

5.1 Normal form of the same magnetic cantilever

In the dimensionless chart of Section 2.3, write

$$\begin{aligned} h_\star &= h_2 + g(d^\top X), \\ h_2 &= \frac{1}{2}\sum_{j=1}^{n} \nu_j (X_j^2 + Y_j^2), \\ g(z) &= 1 - \cos z - \frac{z^2}{2}. \end{aligned} \tag{57}$$

The magnetic term in this expression is the complete analytic function. In particular, its fourth-order part and its untruncated remainder are

$$\begin{aligned} h_{4,\star} &= -\frac{(d^\top X)^4}{24}, \\ T_{6,\star} &= 1 - \cos(d^\top X) - \frac{(d^\top X)^2}{2} \\ &\quad + \frac{(d^\top X)^4}{24}. \end{aligned}$$

Proposition 6. Magnetic twist and analytic tail. *Let $I_j = (X_j^2 + Y_j^2)/2$, and let $A_\star$ denote the Hessian with respect to I of the angular average of $h_{4,\star}$. With $v = (d_1^2, \dots, d_n^2)^\top$,*

$$\begin{aligned} (A_\star)_{jj} &= -\frac{d_j^4}{8}, \\ (A_\star)_{ij} &= -\frac{d_i^2 d_j^2}{4} \quad (i \ne j), \\ A_\star &= \frac{\mathrm{diag}(d_j^4) - 2vv^\top}{8}. \end{aligned} \tag{58}$$

and

$$\det A_\star = (1 - 2n) 8^{-n} \prod_{j=1}^{n} d_j^4. \tag{59}$$

Thus every retained mode with nonzero tip-rotation participation contributes to a nonsingular twist. For $n \ge 2$, the twist has one negative and $n - 1$ positive eigenvalues; positive definiteness is not required here. If $D = \sum_j |d_j|$ and $\mathbb{D}_r^{2n}$ is a complex Cartesian polydisc, then

$$\|T_{6,\star}\|_r \le \frac{(Dr)^6}{720} \cosh(Dr). \tag{60}$$

Proof. The angular averages are $\langle X_j^4 \rangle = 3I_j^2/2$ and $\langle X_i^2 X_j^2 \rangle = I_i I_j$. All terms with an odd exponent in some X_j have zero average. Substitution gives (58). Factoring $\mathrm{diag}(d_j^2)$ on both sides gives a matrix congruent to $(I_n - 2\mathbf{1}\mathbf{1}^\top)/8$. Its eigenvalues are $1/8$ with multiplicity $n - 1$ and $(1 - 2n)/8$, which proves the determinant and inertia statements. For the analytic tail, expand the cosine and use $(6 + 2k)! \ge 6!(2k)!$ and $|d^\top X| \le Dr$.

For the two-mode, 32-element model, the magnetic-bias modes give $\nu \simeq (1, 6.152117504)$ and $d \simeq (0.22567077, 0.32179881)$. Consequently $\det A_\star \simeq -1.303708 \times 10^{-6}$ and $\sigma_{\min}(A_\star) \simeq 5.8064 \times 10^{-4}$. These quantities belong to the same cantilever model used for learning and shape reconstruction.

5.2 Complete learned Hamiltonian and transfer of twist

The fitted family is

$$h_\eta = h_2 + \frac{b}{s_4}\, g(d^\top X) + \sum_{j=1}^{34} c_j \psi_j(X, Y), \qquad (61)$$

where $\eta = (b, c_1, \ldots, c_{34})$, $a = b/s_4$, $s_4 = \|q_\star\|_2$ is the fixed quartic coefficient normalizer and the ψ_j span the coefficient-space complement of the target quartic direction. The class contains the complete target at $\eta_\star = (s_4, 0, \ldots, 0)$; no cosine truncation error is assigned to the training labels. The polynomial part is represented by matched real or complex features. All bounds use the actually exported feature matrix, so numerical orthogonality of its columns is not a proof assumption.

Let $F(z) = \big(g(d^\top X)/s_4, \psi_1(z), \ldots, \psi_{34}(z)\big)$, and put $D(z) = [\nabla F_0(z), \ldots, \nabla F_{34}(z)] \in \mathbb{R}^{4\times 35}$. With $G_N = (4N)^{-1} \sum_i D(z_i)^\top D(z_i)$, $\lambda_{\min}(G_N) \geq \underline{\lambda} > 0$, empirical derivative root-mean-square loss r_N (averaged over all $4N$ scalar observations), and a certified derivative-label error with the same root-mean-square normalization bounded by ϵ_N, the deterministic estimate is

$$\|\eta - \eta_\star\|_2 \leq b_N := \frac{r_N + \epsilon_N}{\sqrt{\underline{\lambda}}}. \qquad (62)$$

This estimate follows from the smallest singular value of the stacked derivative design matrix and the triangle inequality.

The effective quartic coefficient vector is $q_\eta = (b/s_4)q_\star + Uc$, where U is the exported matrix of polynomial features. Let $\mathcal{A}q$ be the symmetric twist matrix obtained by angular averaging. Then the directly computable bound

$$\|A_\eta - A_\star\|_2 \leq \|\mathcal{A}[q_\star/s_4\, U]\|_{2\to F}\, b_N + \delta_{A,\mathrm{dec}} < \sigma_{\min}(A_\star) \qquad (63)$$

is sufficient for nonsingularity of the trained twist. The subscript F denotes the Frobenius norm on symmetric matrices. In the two-mode raw quartic basis, the previously derived bound $\|\mathcal{A}\Delta q\|_2 \leq \sqrt{19}\|\Delta q\|_2$ is also valid. A direct rational or interval check of A_η can be sharper than the sufficient loss-based bound. Here $\delta_{A,\mathrm{dec}}$ bounds the exported decoder's rounding error relative to its exact coefficient map; it is zero for exact arithmetic and is explicitly included in the numerical certificate. Specifically, the exported effective-quartic coefficient error δ_{dec} gives $\delta_{A,\mathrm{dec}} = \sqrt{19}\delta_{\mathrm{dec}}$.

Define the finite-feature analytic bound by

$$B_F(r)^2 = \left(\frac{(Dr)^4}{24 s_4}\cosh(Dr)\right)^2 + r^8 \sum_{j=1}^{34} \|\psi_j\|^2_{\mathrm{coeff},1}.$$

The same coefficient estimate gives a complete analytic perturbation bound,

$$\|h_\eta - h_\star\|_r \leq b_N B_F(r) + \delta_{h,\mathrm{dec}}(r). \qquad (64)$$

This controls the entire learned function on a complex domain, including every higher-order magnetic term.

Suppose $\alpha_4 = \min_{0<|k|_1\leq 4} |k\cdot\nu| > 0$. The exact quartic Lie transformation associated with the full effective $h_{4,\eta}$ gives

$$h_\eta \circ \Phi_\eta = \nu\cdot I + \tfrac{1}{2} I^\top A_\eta I + \mathcal{R}_{6,\eta}.$$

If C_4, C_Z, C_χ are the Cartesian coefficient norms of its quartic part, angular average and homological generator, then

$$\|\mathcal{R}_{6,\eta}\|_r \leq 512 C_\chi (C_4 + C_Z) r^6 + \frac{|a|(2Dr)^6}{720}\cosh(2Dr). \qquad (65)$$

This holds when $32 C_\chi r^2 < 1$. The first term is the exact quartic Lie remainder; the second bounds $aT_{6,\star}\circ\Phi_\eta$ because the time-one flow maps the radius-r polydisc into the radius-$2r$ polydisc. This identity retains the full cosine rather than applying a theorem to an isolated quartic approximation.

With nonsingular A_η and fourth-order nonresonance, the ordinary Kolmogorov theorem applies on sufficiently small positive-action boxes (Pöschel 2001). Indeed, their action width is $O(r^2)$, their integrable Hessian is A_η, and (65) is $O(r^6)$. Choosing a Diophantine constant proportional to r^2 makes the analytic KAM smallness quotient $O(r^2)$; reducing its proportionality constant first leaves a positive-measure set of admissible frequencies. This establishes local KAM families for the complete target and trained cantilever Hamiltonians. The finite-amplitude result below is verified separately by an a posteriori theorem and is not inferred from this asymptotic statement.

5.3 Validated invariant tori at finite physical amplitude

The local normal-form argument can be supplemented by a finite-amplitude certificate for the same complete Hamiltonian. We use the a posteriori theorem of Haro and

Luque (2019); the equation and table numbering below refer to Theorem 3.1 and Appendix B, Tables 1–3, of its archived preprint, arXiv:1711.05515v1. The computation includes the analytic interpolation error and uses ball arithmetic, rather than identifying a small residual at finitely many sample points with an analytic norm.

The mathematical model certified here is specified exactly by the exported binary coefficients. In particular,

$$\begin{aligned} \nu &= \left(1, \frac{6926668524826847}{1125899906842624}\right), \\ d &= \left(\frac{4065323231059343}{18014398509481984}, \frac{2898505962069547}{9007199254740992}\right). \end{aligned} \tag{66}$$

The target uses $a = 1$ and zero quartic residual in (61). Each trained case uses the exact dyadic value of its exported cosine multiplier and every exported quartic coefficient. This convention identifies the function being certified without hiding floating-point parameter uncertainty. It establishes a result for the exported two-mode reduction; independent finite-element and COMSOL comparisons do not themselves prove an infinite-dimensional KAM theorem for the continuum beam.

Theorem 7. Finite-amplitude tori of the target and trained models. *For the target Hamiltonian (57), with parameters (66), for each of the three primary exported GC Hamiltonians with training seeds 17, 43, and 91, and for the additional seed-17 fit at noise amplitude 10^{-8}, and for the GC model fitted to independent COMSOL position–velocity trajectories, there is a real analytic, embedded Lagrangian torus $K_\infty : \mathbb{T}^2 \to \mathbb{R}^4$ satisfying the invariance equation below, with the displayed frequency for the first five models:*

$$\begin{aligned} X_h(K_\infty) &= DK_\infty \varpi, \\ \varpi &= \frac{w_1}{2\pi}(1, \zeta), \\ w_1 &= \frac{49999589341}{50000000000}. \end{aligned} \tag{67}$$

where, with $\phi = (1+\sqrt{5})/2$,

$$\zeta = \frac{1876278\phi + 592877}{304979\phi + 96369}. \tag{68}$$

For the COMSOL-fitted model, the frequency is instead specified by

$$\begin{aligned} w_1 &= \frac{99999143141}{100000000000}, \\ \zeta &= \frac{974993\phi + 427310}{158480\phi + 69457}. \end{aligned} \tag{69}$$

It is chosen near that actual model's quartic normal-form frequency. Angles have period one; ϖ is measured in cycles per unit dimensionless time. Each torus lies within 3×10^{-41}, in the supremum norm on the complex strip of width $1/80$, of its delivered finite Fourier polynomial. The reconstructed beam has nonzero millimetre-scale motion: the certified tip-displacement witness and the bound valid for every point of the centerline are given in Table 2.

Proof. We give the analytic estimates and specify the finite checks that close them. Vector norms are maximum norms. Following the source convention, analytic matrix norms are $\max_i \sum_j \sup_\theta |M_{ij}(\theta)|$, with componentwise suprema on $\mathbb{T}^2_\rho = \{\theta : |\operatorname{Im}\theta|_\infty < \rho\}$. The phase-space domain is $\mathbb{D}^4_{3/10}$.

Frequency and the cohomological equation. The continued fraction of ζ has prefix $[6; 6, 1, 1, 2, 1, 23, 1, 13, 6, 3]$ followed by an infinite tail of ones. For the COMSOL-fitted model the prefix is $[6; 6, 1, 1, 2, 1, 20, 1, 1, 4, 1, 1, 3, 2]$, also followed by ones. All partial quotients in either case are at most 23. The bounded-partial-quotient inequality gives $|q\zeta - p| \geq 1/(25|q|)$ for $q \neq 0$. As $w_1 > 9/10$ and $2\pi < 7$, this implies

$$|k \cdot \varpi| \geq \frac{\gamma}{|k|_1}, \qquad \gamma = \frac{9}{1750}, \qquad k \in \mathbb{Z}^2 \setminus \{0\}. \tag{70}$$

The case $k_2 = 0$ follows directly from $\varpi_1 > 9/70$. We use the weaker exponent $\tau_K = 3$ in the a posteriori theorem. For a zero-mean analytic function v, Fourier division by $2\pi i k \cdot \varpi$ and the count $\#\{k : |k|_1 = j\} = 4j$ give, with $q = e^{-2\pi\delta}$,

$$\begin{aligned} \|\mathcal{R}_\varpi v\|_{\rho-\delta} &\leq \frac{2}{\pi\gamma}\frac{q(1+q)}{(1-q)^3}\|v\|_\rho \\ &\leq \frac{e^{4\pi\rho}}{2\pi^4\gamma\delta^3}\|v\|_\rho < \frac{\|v\|_\rho}{\gamma\delta^3}. \end{aligned}$$

These inequalities hold for $0 < \delta < \rho = 1/40$, so the cohomological constant $c_R = 1$ is justified directly. The verification reconstructs the convergent matrix of the continued fraction and checks these metadata, rather than assuming that an arbitrary numerical frequency is Diophantine.

Global bounds and theorem constants. Write $h = h_2 + ag(d^\top X) + p_4$, with quartic coefficient norm $C_4 = \sum_\alpha |(p_4)_\alpha|$. The exact exported coefficients satisfy $|a| \leq 2$, $C_4 < 1/100$, $\|d\|_1 < 11/20$, and $\max \nu_j < 7$. The full cosine and its derivatives then give, on $\mathbb{D}^4_{3/10}$,

$$\begin{aligned} c_{X_h,0} &= 3, \quad c_{X_h,1} = 8, \\ c_{X_h^\top,1} &= 32, \quad c_{X_h,2} = 1. \end{aligned} \tag{71}$$

The third quantity bounds the derivative of the *row-vector map* $X_h^\top$. It bounds the sum of the absolute Jacobian

entries and is not the matrix norm of $(DX_h)^\top$. For example, its linear part is $2(\nu_1+\nu_2)<28$; its remaining bound is $2\|d\|_1^2(\cosh(3\|d\|_1/10)-1)+12C_4(3/10)^2<1$. For the second derivative, the row-sum tensor estimate $2\|d\|_1^3\sinh(3\|d\|_1/10)+24C_4(3/10)<1$ suffices. All geometric tensors are constant canonical tensors with norm one and zero derivatives.

Let $J=\left(\begin{smallmatrix}0 & I\\ -I & 0\end{smallmatrix}\right)$, $L=DK$, $B=(L^\top L)^{-1}$, and $N=-JLB$. In the source convention the symplectic matrix is $\Omega=-J$. The torsion used in the theorem is therefore

$$T=-N^\top J\big(DX_h(K)N-D_\varpi N\big). \tag{72}$$

There are no additional first integrals ($n=d=2$, canonical Case III). Choose $\rho=1/40$, $\rho_\infty=1/80$, $a_1=a_2=2$, $a_3=12$, $\delta=\rho/12$, and the iteration parameter $\nu_{\mathrm{it}}=1$. The bounds used with strict margins are

$$\begin{gathered}\|DK\|_\rho<1<\sigma_K=2,\\ \|DK^\top\|_\rho<\tfrac{3}{2}<\sigma_{K^\top}=2,\\ \|B\|_\rho<3<\sigma_B=4,\end{gathered}$$

$$\|\langle T\rangle^{-1}\|<10^5<\sigma_T=2\times10^5,\qquad \|K\|_\rho<\tfrac{1}{5}.$$

The last inequality leaves a phase-space distance margin $1/10$. Substitution in Tables 1–3 gives exact rational constants, delivered with the certificate; their decimal displays are

$$\begin{aligned}C_1&\simeq 4.843521623\times10^{46},\\ C_2&\simeq 4.940319879\times10^{21},\\ \frac{\gamma^4\rho^{12}}{C_1}&>8.6086\times10^{-76}.\end{aligned} \tag{73}$$

The rational recurrence is also provided as an executable transcription of those tables; all decisions use rational or ball bounds, not rounded decimal displays. The source's C_{sym} is evaluated from its Eq. (4.39), giving $\sigma_B^2C_{\Omega_L}=192$.

Analytic residual and torsion checks. For each case the delivered K is an exact dyadic finite Fourier polynomial supported on $|k_j|\le63$. Conjugate symmetry is checked exactly. Its derivative and wide-strip norms are bounded by $\sum_k|K_k|e^{2\pi s|k|_1}$ and the corresponding differentiated sums. To bound B, compare K with $K_0=(\frac{1}{10}\cos2\pi\theta_1,\frac{1}{10}\cos2\pi\theta_2,-\frac{1}{10}\sin2\pi\theta_1,-\frac{1}{10}\sin2\pi\theta_2)$. Put $e_L=\|D(K-K_0)\|$ and $e_{L^\top}=\|D(K-K_0)^\top\|$. As $DK_0^\top DK_0=(2\pi/10)^2I$ exactly, the checked inequality

$$\begin{aligned}&\|DK_0^\top\|e_L+e_{L^\top}\|DK_0\|\\ &\quad+e_{L^\top}e_L<(2\pi/10)^2\end{aligned}$$

provides a Neumann-series inverse bound for the analytic Gram matrix.

The residual $E=X_h(K)-DK\varpi$ is evaluated with 512-bit Arb balls on a 256×256 grid, including ball sine evaluations and all quartic terms. Let $\mathcal{I}_mE$ be its trigonometric interpolant, $\widehat{\rho}=1/4$, and $M\ge\|E\|_{\widehat{\rho}}$, computed directly from the finite Fourier polynomial and the entire Hamiltonian. Fourier decay, the two-dimensional union bound for the omitted modes, and the alias sum give

$$\begin{aligned}\|E\|_\rho&\le\|\mathcal{I}_mE\|_\rho+E_{\mathrm{tail}},\\ E_{\mathrm{tail}}&=8M\frac{1+q}{(1-q)^2}e^{-\pi(\widehat{\rho}-\rho)m}.\end{aligned} \tag{74}$$

Here $q=e^{-2\pi(\widehat{\rho}-\rho)}$ and $m=256$. Both the omitted Fourier tail and the aliased coefficients are included in the factor 8. The larger strip is used only to estimate the composition residual; the entire full-cosine Hamiltonian is defined there even when $K(\mathbb{T}^2_{\widehat{\rho}})$ leaves the smaller phase-space domain used by the KAM iteration.

For the mean torsion, the same Gram estimate on the strip $7/100$ proves $\|N\|,\|N^\top\|<6$. Since $\|\varpi\|_1<2$, Cauchy's inequality on the strip $6/100$ gives $\|D_\varpi N\|<1200$, and hence $\|T\|<6(8\cdot6+1200)<10^4$. An Arb trapezoidal average on 128^2 nodes is corrected by

$$10^4\left[\left(\frac{1+z}{1-z}\right)^2-1\right],\qquad z=e^{-2\pi(6/100)128}.$$

The derivative of B in this evaluation is calculated by differentiating its inverse identity, so no differentiated interpolation of N is used. Inverting the resulting interval 2×2 matrix proves the stated bound on $\langle T\rangle^{-1}$.

The resulting bounds in Table 2 satisfy

$$\frac{C_1\|E\|_\rho}{\gamma^4\rho^{12}}<1. \tag{75}$$

The cited theorem supplies the invariant Lagrangian torus and $\|K_\infty-K\|_{\rho_\infty}\le C_2\|E\|_\rho/(\gamma^2\rho^6)<3\times10^{-41}$. For completeness, embeddedness can also be checked quantitatively: writing $X_j-iY_j=0.1e^{2\pi i\theta_j}+\xi_j$, the verified C^0 and C^1 differences from K_0 make the phase map a degree-one map with derivative less than $1/2$ from the identity. It is a diffeomorphism. The same inequality holds after adding the KAM correction and its Cauchy derivative bound on a smaller strip.

Physical displacement. The exact exported shape matrix maps X to the Hermite nodal displacements and slopes. On an element of length ℓ, the cubic Bernstein control values are w_0, $w_0+\ell\vartheta_0/3$, $w_1-\ell\vartheta_1/3$, and w_1. For a

control row b, its value on the invariant torus is bounded by $\sum_{j=1}^{2} |b_j|(\sum_k |(X_j)_k| + 3 \times 10^{-41})$. The convex-hull property therefore bounds every centerline point, every phase, and all time. A ball evaluation at phase $(0, 0)$ gives the separate positive lower witness in the table. These bounds use the same exported shape map as the learning and beam reconstruction.

The additional low-noise fit connects the two certificates on one actual learned Hamiltonian. Its nonzero derivative MSE is $3.3866649 \times 10^{-17}$. The outward Gram lower bound 6.0167×10^{-8}, together with the label and decoder error budgets, gives a twist-error upper bound 2.81119×10^{-4}, smaller than the target singular-value lower bound 3.03081×10^{-4}. Its saved coefficients are exactly those used by the fifth row of the table. Thus the nonzero-loss twist certificate and the finite-amplitude KAM certificate concern the same fitted function; the latter additionally uses the full analytic invariance residual, as required by the theorem.

The sixth row certifies the actual GC function obtained from four independent COMSOL position–velocity trajectories by integral matching. The fit uses state increments and feature integrals, without reconstructed acceleration labels. Its exported cosine multiplier is $a = 1.022324882191665$; every exported quartic term is included in the analytic verification. The coefficient file and the four source trajectory hashes are linked in the proof manifest. The frequency in (69) admits the same conservative $\gamma = 9/1750$, so the global bounds and theorem constants above remain unchanged. In this case the checked inverse averaged torsion is below 94762, within the required 10^5 margin.

This result certifies one specified nonresonant torus in each Hamiltonian at a useful physical amplitude. It does not attach a measured phase-space volume to these individual tori. Positive-measure families near the elliptic equilibrium follow from the separate normal-form argument above; the finite-amplitude theorem and the energy-based safety theorem have distinct hypotheses and conclusions.

5.4 Positive-measure families around the certified tori

The nondegeneracy retained by the a posteriori iteration gives a further conclusion at the same finite-amplitude motion.

Corollary 8. KAM stability of the certified finite-amplitude tori. *Each of the six tori in Theorem 7 is accumulated by positive-measure families of analytic Lagrangian invariant tori. In analytic symplectic tubular coordinates (θ, I) centered on that torus, their relative Lebesgue measure in $\mathbb{T}^2 \times \{|I|_\infty < r\}$ tends to one as $r \to 0$. This is an asymptotic statement about neighborhoods of the certified millimetre-amplitude torus; it does not specify a numerical tube thickness or a percentage in a preassigned experimental domain.*

Proof. The inequalities controlling the inverse averaged torsion in the Haro–Luque iteration are uniform at every step: see their Eqs. (5.65)–(5.67) and the summation in Section 5.3. The strict margin used in (73) therefore gives

$$\|\langle T_\infty \rangle^{-1}\| \le 2 \times 10^5.$$

This is nondegeneracy of the final invariant object, not only of its numerical approximation.

An analytic Lagrangian neighborhood chart Ψ can be chosen with $\Psi(\theta, 0) = K_\infty(\theta)$ and first derivative $D\Psi(\theta, 0) = [L_\infty, N_\infty]$. One construction starts with $K_\infty(\theta) + N_\infty(\theta)I$ and uses the relative analytic Moser correction. The forms agree on the zero section; the relative primitive and its correcting vector field are $O(|I|^2)$, so the correction preserves this first jet. In these coordinates invariance implies

$$\begin{aligned} h \circ \Psi = e + \varpi \cdot I + \tfrac{1}{2} I^\top A(\theta) I \\ + O(|I|^3). \end{aligned} \tag{76}$$

The automatic-reducibility identity of Haro–Luque Section 2.4 is

$$\begin{aligned} DX_h(K_\infty)P - D_\varpi P \\ = P \begin{pmatrix} 0 & T_\infty \\ 0 & 0 \end{pmatrix}, \end{aligned}$$

where $P = [L_\infty, N_\infty]$. Comparing this with the linearized vector field of (76) gives $A = T_\infty$. More generally, changing the symplectic normal frame to $N_\infty + L_\infty S(\theta)$, with S symmetric, changes this block to $T_\infty - D_\varpi S$, whose average is the same. Hence $\overline{A} = \langle A \rangle$ is nonsingular independently of this choice of normal frame.

The quadratic normal form is justified without a formal-series convergence assumption. Solve the matrix homological equation $D_\varpi S = A - \overline{A}$ with zero average by

$$S_k = \frac{A_k}{2\pi i k \cdot \varpi} \quad (k \neq 0), \qquad S_0 = 0.$$

Equation (70) and analyticity make this series convergent on every strictly smaller angle strip; the cohomological bound established above controls it there. The exact time-one Hamiltonian flow of $\chi_2(\theta, I) = \frac{1}{2} I^\top S(\theta) I$ exists on a sufficiently thin tube and fixes $I = 0$. With the pullback convention $L_\chi F = \{F, \chi\}$, it changes the quadratic coefficient to $\overline{A}$ and leaves an analytic $O(|I|^3)$ remainder.

Table 2. Finite-amplitude certificates for the complete target and actual trained Hamiltonians. Displayed bounds are rounded outward. The lower witness is tip displacement at one phase; the upper bound covers the entire reconstructed centerline at all phases.

Model	$\|E\|_\rho$ upper	KAM quotient upper	Tip lower (mm)	Centerline upper (mm)
Target	$2.7310\,10^{-77}$	0.031724	1.8977905	1.8977964
GC17	$2.7372\,10^{-77}$	0.031796	1.8927460	1.8927517
GC43	$2.7538\,10^{-77}$	0.031988	1.8837514	1.8837573
GC91	$2.7285\,10^{-77}$	0.031694	1.9077295	1.9077354
GC17 (10^{-8} noise)	$2.7311\,10^{-77}$	0.031724	1.8977399	1.8977457
GC (COMSOL states)	$2.7330\,10^{-77}$	0.031746	1.8977374	1.8977433

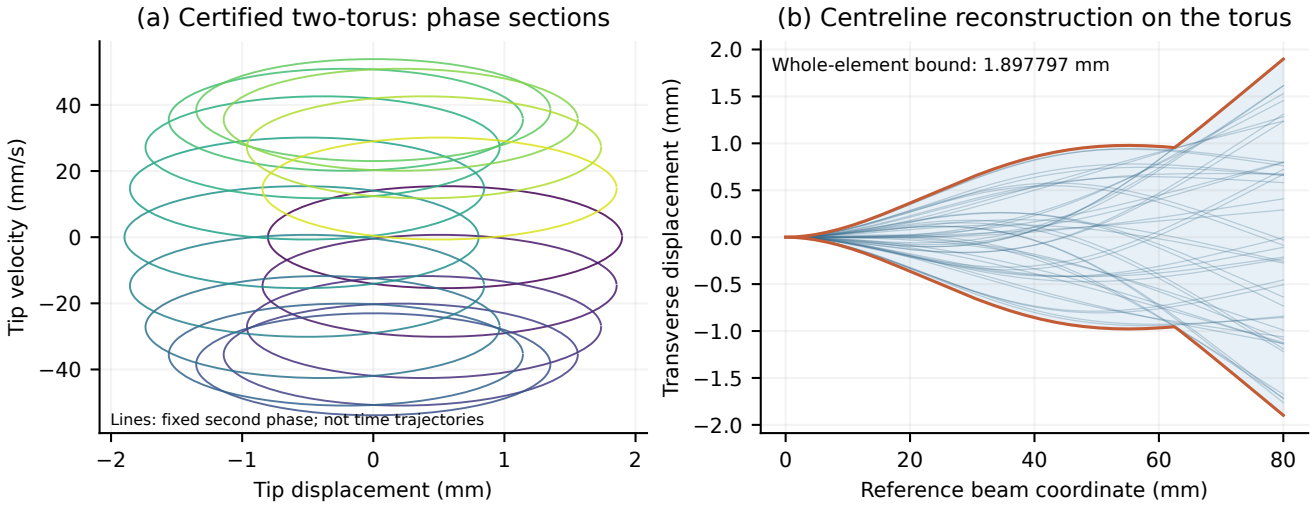


Figure 1. The certified target torus in the same SI beam coordinates. (a) Tip displacement and velocity along sampled sections with the second phase fixed; these curves are not time trajectories. (b) Sampled centerlines and a plotted phase envelope. The separate whole-element Bernstein/Fourier certificate bounds every centerline point on the corrected invariant torus by 1.897797 mm. Sampling illustrates the geometry; the analytic certificate supplies the all-phase bound.

The resulting normal form satisfies the hypotheses of Bounemoura (2016, Theorem 6, analytic case): the frequency is nonresonant, the averaged quadratic form is nonsingular, and the full remainder is analytic. That theorem gives the claimed density-one families. Symplectic coordinate changes preserve volume, and the constant factor relating the dimensionless chart to SI canonical coordinates cancels in relative measure.

Obtaining a numerical percentage for a fixed tube would additionally require bounds for the chart radius, its full analytic remainder and the constants in the family theorem. The present finite-amplitude certificate gives explicit motion amplitudes; this corollary supplies the neighborhood-family conclusion without assigning an uncomputed thickness to it.

6 Reproducible learning and independent COMSOL validation

All experiments in this section use the clamped-free beam, static-field eigenvectors, complete magnetic cosine, and SI parameters defined above. The two-mode learning problem and its higher-order validation models therefore share the same mechanics. The reference energy is

$$h_\star(X,Y) = \frac{1}{2}\sum_{j=1}^{2}\nu_j(X_j^2+Y_j^2) + g(d^TX),$$
$$g(s) = 1-\cos s-\frac{1}{2}s^2. \tag{77}$$

Here $\nu = (1, 6.152117504)$ and $d = (0.225670773, 0.321798806)^T$; one dimensionless energy unit is 6.0×10^{-7} J and $\tau = 11.417287283t$. The tip displacement in metres is $0.013506054X_1 + 0.005470926X_2$. These mappings are used for every reported displacement, slope, and energy error, including the independent COMSOL comparison.

6.1 Finite-feature identification and matched controls

The CHNN uses $z_j = (X_j + \mathrm{i}Y_j)/\sqrt{2}$, conjugate polynomial features, and a complex exponential feature for the full cosine. Its real energy readout is

$$h_\vartheta = h_2 + a\,g(d^TX) + m_4(X,Y)^TUc, \tag{78}$$

where m_4 lists the 35 degree-four real monomials in (X_1, X_2, Y_1, Y_2). If v_4 is the coefficient vector of $-(d^T X)^4/24$, the 34 columns of U span its orthogonal complement. The feature $g/\|v_4\|_2$ and these complementary quartics give 35 fitted real parameters, avoid duplicating the leading cosine term, and retain the complete analytic cosine tail. The target corresponds to $a = 1, c = 0$ and is therefore in the feature class. The stored matrix U, rather than idealized exact orthogonality, is used in the certificate computations.

We compare real (R) and complex (C) implementations, each with either the fixed mechanical quadratic part (G) or a fitted general quadratic part (B). Thus GR and GC have 35 fitted real parameters, whereas BR and BC have 45. The two representations span the same real energy class, receive identical data, and use the same regularizer expressed in common real feature coordinates. They are a controlled representation comparison: complex arithmetic is actually evaluated, but an intrinsic complex-valued accuracy advantage is not expected from an invertible change of features.

For each seed in $\{17, 43, 91\}$, we generate 12 conservative trajectories of 4 s, sampled at 129 times. Initial modal radii are drawn uniformly from $[0.03, 0.20]$ and $[0.03, 0.16]$, with independent uniform phases. The largest training tip slope is 0.0891 rad. Each component of the dimensionless vector-field label receives independent uniform noise in $[-10^{-6}, 10^{-6}]$; states are noiseless. Linear readout fitting minimizes derivative mean-square error with a 10^{-14} quadratic regularizer. Eight independent trajectories, generated with a separate seed and integrated for 16 s at 1601 output times, are held out in full. The reference solver is DOP853 with relative and absolute tolerances 2×10^{-10} and 2×10^{-12}. Tightening both tolerances by a factor of 100 changes the tip of the first held-out trajectory by at most 1.92×10^{-8} mm.

Table 3 and Fig. 2 show the benefit of fixing the known quadratic mechanics in this experiment, while matched real and complex fits agree to numerical precision. GC identifies $a = 1.00326 \pm 0.00154$ with nonzero training residuals. The energy error evaluates the true energy along the predicted and reference states; it is not an energy-conservation surrogate for trajectory accuracy. The small fitting errors reflect exact representation, accurate synthetic derivatives, and deliberately controlled label noise. They do not establish corresponding physical measurement accuracy.

We also continue one independent initial condition for 200 s, or 363 first-mode linear periods, using the three primary GC fits. The seed-level tip RMSE ranges from 1.38×10^{-4} to 4.03×10^{-4} mm; the largest modal phase error is 4.19×10^{-4} rad. The relative drift of each learned model's own energy remains below 5.11×10^{-9} with conservative integration at relative and absolute tolerances 2×10^{-11} and 2×10^{-13}. Figure 3 shows its Poincaré section, phase error, and numerical energy drift. These finite-time diagnostics assess the fitted dynamics; they do not estimate the measure of KAM tori or replace the analytic certificate.

6.2 A nonzero-loss certificate for the fitted magnetic energy

The feature derivatives make the empirical Gram matrix and the complete quartic Taylor map computable. We repeat the same three fits at noise amplitudes 10^{-8} and 10^{-9}, preserving the original 10^{-6} experiments. Outward-rounded interval calculations enclose the derivatives of the analytic sine feature at the stored states, the Gram inverse residual, the observed loss, clean-label rounding, and the conversion from readout weights to the exported energy coefficients. The degree-17 sine series is enclosed with its degree-19 remainder.

For the target magnetic quartic, $A_{jj} = -d_j^4/8$ and $A_{12} = -d_1^2 d_2^2/4$. This twist is indefinite and nonsingular. Exact rational arithmetic on the stored dyadic d gives the conservative lower bound $\sigma_{\min}(A_\star) > 3.03081 \times 10^{-4}$. If G is the derivative Gram matrix, B the quartic coefficient map, ℓ the noisy derivative MSE, and η_{eff} the enclosed noise and arithmetic budget, the verified sufficient inequality is

$$\sqrt{19}\left(\|B\|_2 \frac{\sqrt{\ell} + \eta_{\text{eff}}}{\sqrt{\lambda_{\min}(G)}} + \delta_{\text{dec}}\right) < \sigma_{\min}(A_\star), \qquad (79)$$

where δ_{dec} encloses readout decoding roundoff. At $\eta = 10^{-8}$, Table 4 verifies this inequality with strictly positive losses for all seeds. Thus the data-dependent error bound itself transfers nondegeneracy to the actual learned Hamiltonians. The local KAM conclusions additionally use the nonresonance and analytic normal-form hypotheses proved for this same model. This table alone does not certify a finite experimental oscillation radius. For the primary 10^{-6} fits, this sufficient error bound is too loose; their directly computed twist singular values are nevertheless between 5.78×10^{-4} and 5.93×10^{-4}.

6.3 Learning dissipation and opening the magnetic input port

The imaginary readout is the nonnegative Rayleigh potential $\mathcal{D}_\gamma(v) = \gamma \sum_j v_j^2/(2\nu_j)$, with $v = h_{\vartheta,Y}$. For each seed, eight additional damped, unforced 5 s trajectories provide

Table 3. Conservative held-out errors, mean ± sample standard deviation over three training seeds. Each seed is evaluated on all eight independent 16 s trajectories. These are errors within the declared simulated model.

Model	Tip RMSE (mm)	Slope RMSE (mrad)	Energy RMSE (nJ)
BR	$(2.28\pm0.75)10^{-5}$	$(5.80\pm1.20)10^{-4}$	$(3.21\pm0.51)10^{-6}$
BC	$(2.28\pm0.75)10^{-5}$	$(5.80\pm1.20)10^{-4}$	$(3.21\pm0.51)10^{-6}$
GR	$(1.20\pm0.71)10^{-5}$	$(3.90\pm1.63)10^{-4}$	$(3.14\pm0.71)10^{-6}$
GC	$(1.20\pm0.71)10^{-5}$	$(3.90\pm1.63)10^{-4}$	$(3.14\pm0.71)10^{-6}$

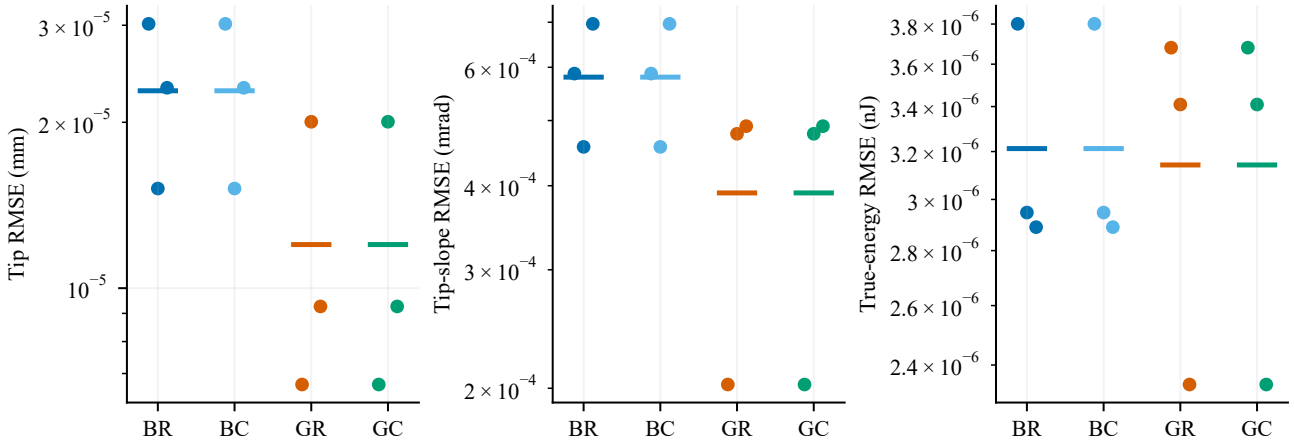


Figure 2. Identical full-cosine beam data and real parameter budgets for the four learning controls. Points are individual seeds and horizontal marks are means. The quadratic mechanics prior changes the result; the matched real/complex representation changes it only at numerical precision.

noisy derivative labels. Nonnegative least squares fits the single γ using the already fitted conservative gradient. In particular, the decoder retains the learned momentum-dependent quartics:

$$\begin{aligned} X' &= h_{\vartheta,Y}, \\ Y' &= -h_{\vartheta,X} - \gamma\,\mathrm{diag}(\nu^{-1})h_{\vartheta,Y} \\ &\quad + u\,d\cos(d^T X). \end{aligned} \tag{80}$$

The fitted dimensional mass-drag coefficients are 0.555585710, 0.555587102, and 0.555584024 s^{-1}, compared with 0.555585237 s^{-1} for the generator. On six independent driven trajectories with amplitudes 4–8 mT and frequency ratios 0.65–1.20 relative to the biased first frequency, the three seed-level tip RMSEs are 8.83×10^{-5}, 1.37×10^{-4}, and 6.40×10^{-5} mm. Closing the learned drag port increases the tip RMSE to approximately 3.01 mm. The integrated work balance is

$$\begin{aligned} h_\vartheta(\tau) - h_\vartheta(0) = &\int_0^\tau u\cos(d^T X) d^T h_{\vartheta,Y}\,ds \\ &- \int_0^\tau \gamma \sum_j h_{\vartheta,Y_j}^2/\nu_j\,ds. \end{aligned} \tag{81}$$

Its largest numerical residual across these cases is 4.41×10^{-17} J. These driven tests assess the port model, rather than autonomous KAM persistence. Some stress cases exceed the channel clearance and are free-motion continuations; safe operating trajectories are evaluated separately with the physical energy bound and input governor.

6.4 Independent COMSOL assembly and nonlinear time integration

COMSOL Multiphysics 6.3 independently assembles the same 32-element clamped-free beam and solves its nonlinear magnetic-tip response. The relative Frobenius differences between the exported and Python consistent-mass matrices are 7.75×10^{-15} for mass and 3.05×10^{-14} for bending stiffness. The nonlinear held-out COMSOL trajectory starts from rest, uses the same mass drag, and applies

$$B_y(t) = 0.006\,[1-\exp(-(t/0.25)^2)]\sin(\Omega t) \quad \mathrm{T} \tag{82}$$

with $\Omega = 9.4449490256$ rad/s for 8 s, relative tolerance 10^{-7}, and 2001 output samples. The exported nodal displacement, rotation, velocity, and angular velocity are evaluated directly, with no fitting to this driven trajectory. Its peak tip displacement is 3.09032 mm. A separate COMSOL solve with relative tolerance 10^{-8} changes the tip by 4.59×10^{-5} mm RMSE and reduces its discrepancy against the full Python FE solution to 1.03×10^{-5} mm RMSE. The original held-out dataset is preserved for the primary comparison below.

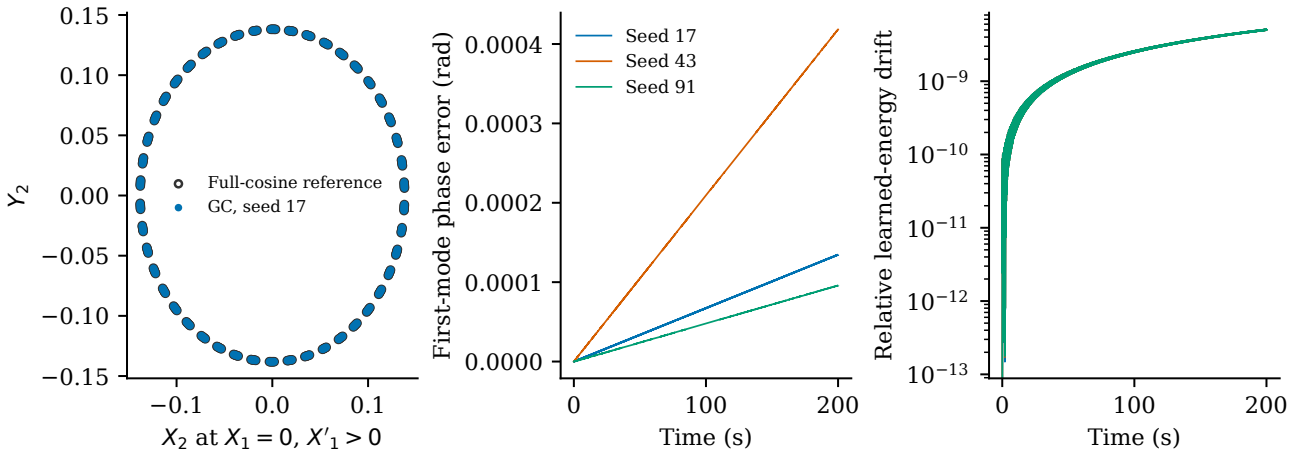


Figure 3. Conservative 200 s continuation of an independent initial state. Left: 364 upward crossings of $X_1 = 0$ for the reference and GC17. Middle: unwrapped first-mode phase errors of the three learned energies. Right: each learned energy's relative integration drift. All ports are closed for these conservative diagnostics.

Table 4. Enclosed loss-to-twist checks at noise amplitude 10^{-8}. All twist-error upper bounds are below 3.03081×10^{-4}.

Seed	Noisy MSE	Gram eigenvalue lower bound	Twist-error upper bound
17	3.387×10^{-17}	6.0167×10^{-8}	2.8112×10^{-4}
43	3.275×10^{-17}	2.2734×10^{-7}	1.4374×10^{-4}
91	3.283×10^{-17}	2.0968×10^{-7}	1.4973×10^{-4}

The nested modal comparison in Table 5 separates reduction error from learning error. Against the exact two-mode response, GC contributes only 1.81×10^{-6}–3.48×10^{-6} mm tip RMSE for this input. Against COMSOL, both have approximately 0.0156 mm RMSE. The two-mode maximum slope discrepancy is 3.69 mrad; its true stored-energy RMSE is 0.132 nJ. These discrepancies decrease with modal enrichment. Figure 5 shows the response and both error scales. The primary training data were generated by the declared Python Galerkin model; the COMSOL time history is an independent held-out validation dataset. Neither dataset is a physical hardware measurement.

6.5 COMSOL-derived training data and an independent integral check

We generate a separate conservative COMSOL dataset comprising four 4 s training trajectories and two independent 8 s test trajectories, sampled every 0.004 s. Initial displacements and velocities excite both retained modes. The remaining 62 modes receive the static response to the nonlinear magnetic residual and its compatible first time derivative; all 64 modes remain active during integration. This initial tail condensation is not assumed to be an exact nonlinear invariant manifold. Native solver-vector checks confirm that the exported initial displacement and velocity are correct; small changes introduced by consistent initialization are handled by starting each predictor from the actual exported state.

The derivative-label construction requires care. At the stored output times, the exported second time derivatives differentiate a time interpolant and show a substantial lag relative to the velocity evolution. For example, the first trajectory has a dimensionless modal equation-residual RMS of 4.58×10^{-2} when those derivatives are used directly. We retain and audit these exports, and instead reconstruct instantaneous acceleration from the exported COMSOL mass and stiffness matrices:

$$\ddot{q} = M_{\mathrm{C}}^{-1} \left[-K_{\mathrm{C}} q - \mu B_0 \sin(\theta_{\mathrm{tip}}) e_\theta \right]. \tag{83}$$

The training states and velocities therefore come from the actual COMSOL time solver, while the acceleration labels use known physical force balance. This is surrogate identification for the declared mechanical model, rather than discovery of an unknown force law or a direct measurement of acceleration. The unresolved-mode force is retained in the audit: its largest component is 3.36×10^{-9} in dimensionless units for the checked training trajectory, and the remaining reconstructed equation-residual RMS is 1.81×10^{-10}.

The additional models are fitted separately and never replace the primary GC coefficients or their certificates. Three seeds vary the additional componentwise label noise, bounded by 10^{-6}, while using the same four COMSOL trajectories. The real/complex and prior/no-prior controls use the same parameter budgets as before.

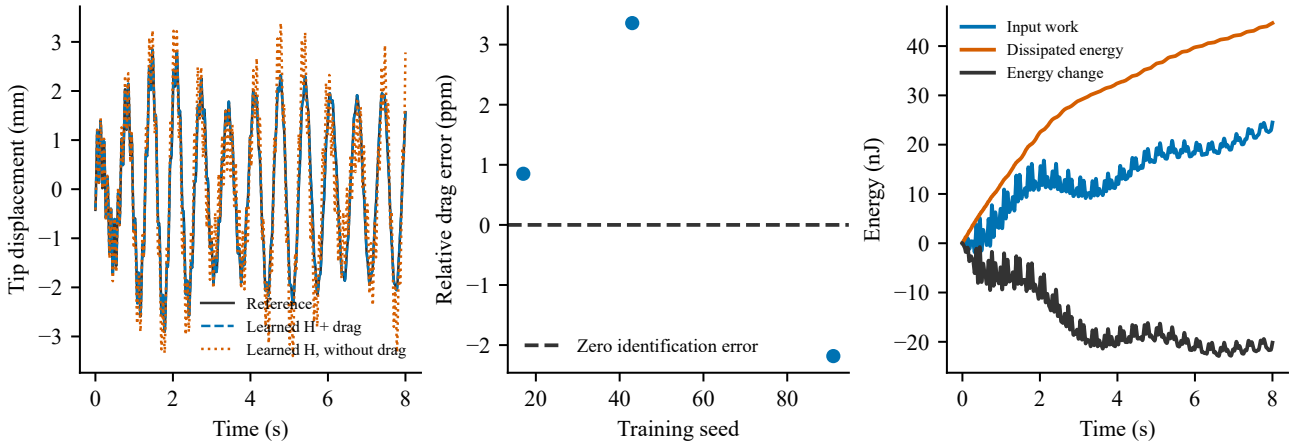


Figure 4. The same learned conservative energy with an identified nonnegative drag readout and magnetic input port. Left: an independent driven trajectory and the drag ablation. Middle: signed relative errors in the drag coefficient. Right: supplied work, dissipated energy, and stored-energy change for that trajectory.

Table 5. Independent nonlinear COMSOL trajectory comparison. Modal models use the common static-field chart. GC includes learned drag; its range covers the three primary training seeds.

Predictor	Tip RMSE (mm)	Maximum tip error (mm)
2-mode exact cosine	0.0156168	0.0226548
4-mode exact cosine	0.00407422	0.00592596
8-mode exact cosine	0.00102484	0.00152782
64-mode Python FE	0.00005614	0.00011835
2-mode learned GC	0.0156171–0.0156177	0.0226545–0.0226549

An independent check uses only the COMSOL displacement and velocity samples. A nine-point closed Newton–Cotes rule integrates the vector-field features on nonoverlapping 0.032 s windows, and the observed state increments provide the regression targets. No acceleration or force-balance label enters this integral fit. With the initial solver settings (relative tolerance 10^{-6} and scaled absolute tolerance 10^{-5}), the integral residual is too large to resolve the weak magnetic correction reliably: on the first trajectory, its RMS is 2.16 times the nonlinear magnetic signal. An unconstrained 35-feature integral fit then yields $a = 1.437$, illustrating that a good fit to a numerically dispersed trajectory need not recover the physical magnetic coefficient. A stricter COMSOL solve, with relative tolerance 10^{-8} and scaled absolute tolerance 10^{-7}, reduces that ratio to 0.112 on the same trajectory. This independent data-quality check prevents small force-balance training losses from concealing inadequate time-integration accuracy.

All six trajectories were subsequently regenerated with these stricter tolerances, while retaining the initial dataset. The successful runs use generalized-alpha integration with high-frequency spectral radius zero and backward-Euler consistent initialization. The four refined training trajectories provide 500 integral windows, or 2000 scalar equations, for the 35 real readout parameters. The resulting GC model has $a = 1.0223248822$ and a nonzero averaged-derivative integral RMS residual of 4.0524×10^{-7}. The corresponding residuals of the quadratic prior alone and the physical full-cosine target are 3.9099×10^{-6} and 6.6454×10^{-7}; the nonlinear magnetic signal RMS is 3.6601×10^{-6}. The actual fitted quartic twist has minimum singular value 5.1166×10^{-4}. The conservative weights were frozen before evaluating the two independent refined test trajectories.

Table 6 and Figure 6 report the held-out errors. The integral fit partly absorbs the remaining time-solver dispersion: its agreement with COMSOL is better than that of the exact closed two-mode target, despite a 2.23% error in the cosine multiplier. These are simulation-identification results and do not imply physical sensor accuracy. We do not infer a rigorous solver-error budget from the requested tolerances or apply the loss-based certificate to these integral residuals.

For comparison, the separately fitted models using the known-force labels in (83) recover $a = 1.0000690$ without added label noise. With the three added-noise seeds, the prior-informed GC models give 0.25347–$0.25940\,\mu$m pooled tip RMSE against the same refined COMSOL tests; their additional error relative to the closed two-mode trajectories is 0.00364–$0.00771\,\mu$m. Matched real and complex features again give coincident results to numerical

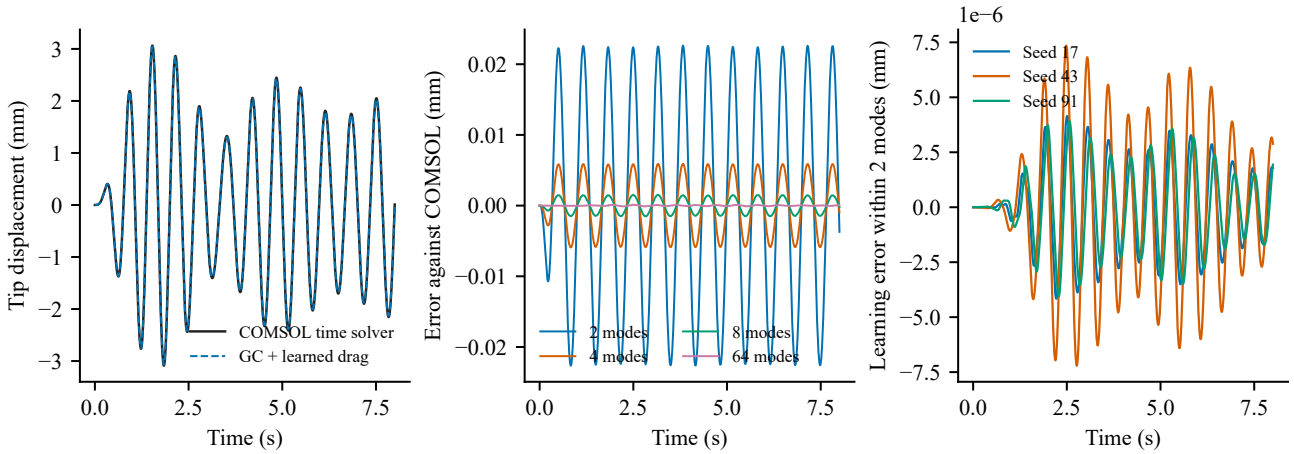


Figure 5. Independent nonlinear COMSOL time integration. Left: COMSOL and the learned two-mode GC response with learned drag. Middle: errors of the nested exact-cosine modal models against COMSOL. Right: the additional GC identification error within the two-mode model. Reduction error is the larger contribution in this validation.

Table 6. Independent refined COMSOL tests of the GC model identified only from displacement and velocity increments. Predictions start from each actual exported initial state. Energy errors use the physical stored-energy function evaluated along the learned trajectory.

Test	Tip RMSE (μm)	Max. tip error (μm)	Slope RMSE (mrad)	Energy RMSE (nJ)	Exact 2-mode tip RMSE (μm)
1	0.14836	0.39240	0.008677	0.0002067	0.24172
2	0.05140	0.17435	0.002411	0.0002253	0.27459

precision. All four model variants, raw derivative exports, reconstructed labels, noise draws, and both solver-precision datasets are retained in the reproducibility package.

Finally, the frozen position–velocity fit is used directly in the independent a posteriori proof of Section 5.3. Its complete cosine and all saved quartic coefficients yield an analytic residual below 2.7330×10^{-77} and a theorem smallness quantity below 0.031746. The resulting exact invariant torus has a tip-displacement witness above 1.8977374 mm and a whole-centreline upper bound below 1.8977433 mm. This certificate concerns the actual fitted Hamiltonian and does not require its coefficients to equal the physical target. The same frozen conservative readout is paired with a nonnegative drag coefficient refitted on the existing Python damped-beam trajectories, $\gamma = 0.0486617702$ for seed 17. Its independent driven COMSOL tip RMSE is 0.015622 mm. Section 7.4 then uses this same energy, full gradient and Hessian, and drag readout to generate the magnetic tracking input. Hashes link the COMSOL training files, frozen weights, KAM certificate, and operational validation.

7 Clearance guarantees and magnetic safety control

7.1 A whole-beam clearance certificate for the same magnetic model

The conservative normal form and the safety calculation use the same clamped-free beam, static field, magnetic dipole, and energy. Safety is a statement about the driven physical equations, with their nonnegative drag; it does not require a driven trajectory to lie on a conservative KAM torus. For the declared planar, small-slope model, write $w(s,t)$ for transverse displacement, $m = \rho A$ for line density, $c_d \geq 0$ for distributed viscous drag, and $\vartheta = w_s(L,t)$ for the tip slope used in the dipole potential. The equations and natural end conditions are

$$m w_{tt} + c_d w_t + EI w_{ssss} = 0, \qquad 0 < s < L, \tag{84}$$

$$w(0,t) = w_s(0,t) = 0, \qquad w_{sss}(L,t) = 0, \tag{85}$$

$$EI w_{ss}(L,t) = \mu\{B_y(t)\cos\vartheta - B_0 \sin\vartheta\}. \tag{86}$$

The end moment is the derivative of the same cosine potential used to generate the training trajectories. In particular, the varying transverse field is an input port, not an additional autonomous Hamiltonian perturbation. For a sufficiently regular solution satisfying the mechanical power identity, the stored energy relative to the aligned static-field equilibrium

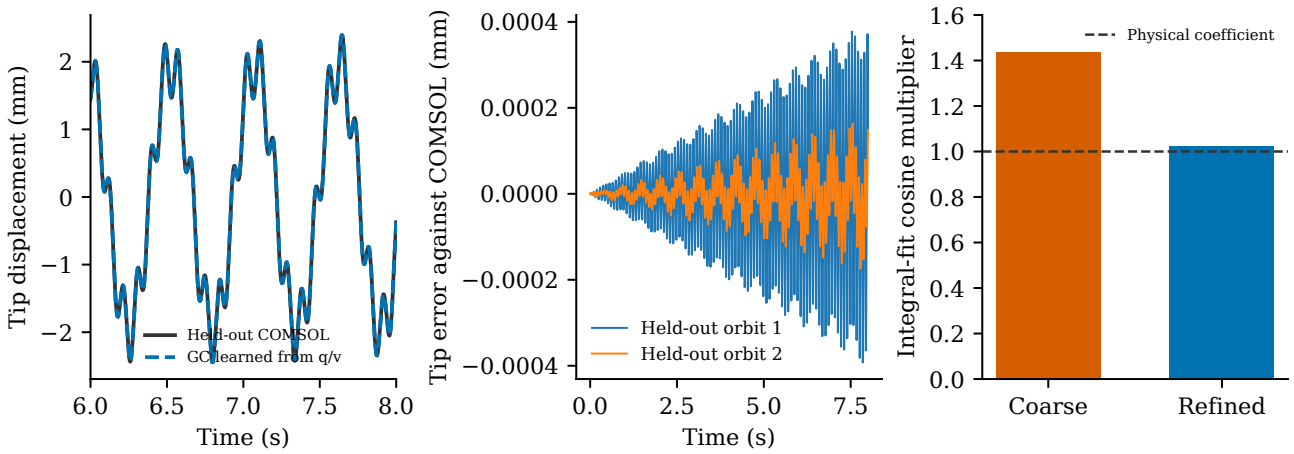


Figure 6. Identification using actual COMSOL displacement and velocity outputs only. Left: independent test response. Middle: prediction errors for both held-out trajectories. Right: the fitted cosine multiplier under the two declared COMSOL tolerance settings; the physical value is one. The refined coefficients are frozen for all subsequent checks.

is

$$E(t) = \frac{1}{2}\int_0^L \{m w_t^2 + EI w_{ss}^2\}\,\mathrm{d}s + \mu B_0(1-\cos\vartheta), \tag{87}$$

$$\dot{E} = -c_d \int_0^L w_t^2\,\mathrm{d}s + \mu B_y \cos\vartheta\,\dot{\vartheta}. \tag{88}$$

All inequalities below also hold for the conforming cubic-Hermite finite element (FE) system and any of its Galerkin subspaces, because their energies are the restrictions of this expression and retain the same input power. In the balanced coordinates of the learning experiment, $E = \mu B_0 h_*$ for the physical target energy, $\vartheta = d^{\mathsf{T}}X$, and the port force in the Y equation is $d(B_y/B_0)\cos(d^{\mathsf{T}}X)$.

Proposition 9. Clearance and field-variation budget. *Assume $B_0 > 0$, $c_d \geq 0$, and an absolutely continuous applied field B_y on $[0,T]$. Define*

$$S_T = \sup_{0\leq t\leq T}|B_y(t)| + |B_y(0)| + \int_0^T |\dot{B}_y(t)|\,\mathrm{d}t, \tag{89}$$

$$C_T = \mu\sqrt{\frac{2L}{EI}}\,S_T, \tag{90}$$

$$R_T = \frac{C_T + \sqrt{C_T^2 + 4E(0)}}{2}. \tag{91}$$

Then, throughout $[0,T]$,

$$\begin{aligned} E(t) &\leq R_T^2, \\ |w(s,t)| &\leq \sqrt{\frac{2s^3}{3EI}}\,R_T, \\ |w_s(s,t)| &\leq \sqrt{\frac{2s}{EI}}\,R_T. \end{aligned} \tag{92}$$

For a straight corridor with wall half-width a and rod radius r, let $c = a - r > 0$. A sufficient condition for positive clearance of the whole rod is $R_T < \sqrt{E_c}$, where

$$E_c = \frac{3EIc^2}{2L^3}. \tag{93}$$

For $E(0) < E_c$, the following directly checkable operating condition is equivalent to this sufficient energy condition:

$$S_T < \frac{E_c - E(0)}{\mu\sqrt{2LE_c/EI}}. \tag{94}$$

These conclusions hold for every nonnegative drag coefficient, including zero.

Proof. Since $B_0 > 0$, the static magnetic energy is nonnegative and $\|w_{ss}\|_{L^2}^2 \leq 2E/EI$. The clamped conditions imply $w(s) = \int_0^s (s-\xi)w_{ss}(\xi)\,\mathrm{d}\xi$ and $w_s(s) = \int_0^s w_{ss}(\xi)\,\mathrm{d}\xi$. Cauchy–Schwarz gives the two spatial inequalities in (92) with $\sqrt{E}$ in place of R_T. Write $D_c(t) = \int_0^t c_d\|w_t\|_{L^2}^2\,\mathrm{d}\tau$. Integrating (88) by parts in the input term gives

$$\begin{aligned} E(t) + D_c(t) - E(0) &= \mu B_y(t)\sin\vartheta(t) \\ &\quad - \mu B_y(0)\sin\vartheta(0) \\ &\quad - \mu\int_0^t \dot{B}_y(\tau)\sin\vartheta(\tau)\,\mathrm{d}\tau. \end{aligned}$$

Set $M_T = \sup_{[0,T]}\sqrt{E(t)}$. Using $|\sin\vartheta| \leq |\vartheta| \leq \sqrt{2L/EI}\,M_T$ and discarding the nonnegative dissipation yields $M_T^2 \leq E(0) + C_T M_T$. The positive root is R_T. The entire radius-r tube about the centerline remains between

the walls when $\sup_s |w(s,t)| < a - r$. Solving $R_T < \sqrt{E_c}$ gives (94).

For a nonuniform planar corridor, let $c(s) > 0$ be the Euclidean distance from the undeformed point $(s, 0)$ to the wall, less the rod radius. The distance-to-wall function is one-Lipschitz, so $|w(s)| < c(s)$ keeps the radius-r tube clear. The same argument replaces E_c by $\inf_{0<s\le L} 3EIc(s)^2/(2s^3)$. For an inclined or curved wall, a local vertical half-width is not a substitute for this Euclidean clearance. This is a sufficient condition for that specified geometry; it is not a vessel-specific claim in the absence of vessel data. The full-continuum inequality removes the need to interpret an observed two-mode reconstruction error as a rigorous neglected-mode bound. It certifies solutions of the stated continuum model with the stated initial-energy and applied-field bounds, rather than claiming an FE trajectory is an exact continuum trajectory.

For the shared SI parameters, choose $a = 4.6$ mm and $r = 0.6$ mm, so that $c = 4$ mm. The resulting E_c is 23.8565 nJ. For a monotone field ramp from zero to A followed by a hold, $S_\infty = 2A$. With zero initial energy, a sufficient condition is

$$A < \frac{\sqrt{3}\,EIc}{4\mu L^2} = 6.8868\ \text{mT}. \tag{95}$$

For $B_y = A\rho(t)\sin(\Omega t)$ with $\rho(t) = 1 - \exp[-(t/0.25\,\text{s})^2]$, $S_T \le A(2 + \Omega T)$, because ρ increases from zero and is at most one. Thus amplitude, frequency and operating duration enter an explicit joint condition by substituting this bound in (94). For $T = 8$ s, $\Omega = 9.444949$ rad/s and $E(0) = 0$, its sufficient amplitude limit is approximately 0.1776 mT. This conservative open-loop test does not certify the default 6 mT sinusoid, even when a simulated trajectory has positive clearance.

7.2 A magnetic-input governor with a full-model energy guarantee

The field-variation test can be strengthened for feedback operation by measuring the actual mechanical input port. Let $\overline{E}_0 \ge E(0)$ be a valid initial-energy bound and integrate

$$W(t) = \overline{E}_0 + \int_0^t \mu B_y \cos\vartheta\,\dot\vartheta\,\mathrm{d}\tau. \tag{96}$$

Equation (88) gives $E(t) \le W(t)$ without measuring or estimating the dissipated energy. Given $E(0) \le \overline{E}_0 < E_{\max} < E_c$, a bounded requested field $|B_{\text{req}}| \le B_{\max}$, and $\kappa > 0$, choose the applied field as the closest value to B_{req} satisfying

$$\begin{aligned} |B_y| &\le B_{\max}, \\ a_p B_y &\le \kappa(E_{\max} - W), \\ a_p &= \mu\cos\vartheta\,\dot\vartheta. \end{aligned} \tag{97}$$

This is a scalar projection onto an interval. Whenever the requested input violates the work constraint and $a_p \neq 0$, the applied field is $B_y = \kappa(E_{\max} - W)/a_p$; otherwise it is unchanged. On $W \le E_{\max}$, zero field is feasible, and this projection never increases the requested field amplitude.

Proposition 10. Port-work safety invariant. *Suppose the actual applied field and actual tip slope and slope rate are available to (96)–(97), and the field command is applied without unbudgeted error or delay. Every closed-loop solution satisfying the mechanical power identity obeys*

$$\begin{aligned} E(t) &\le W(t) \le E_{\max}, \\ \sup_s |w(s,t)| &\le \sqrt{\frac{2L^3 E_{\max}}{3EI}} < c. \end{aligned} \tag{98}$$

Consequently the entire rod retains positive wall clearance for all times on which the stated beam model and port measurements apply, for any $c_d \ge 0$.

Proof. The power identity gives $W - E = \overline{E}_0 - E(0) + \int_0^t c_d\|w_t\|_{L^2}^2\,\mathrm{d}\tau \ge 0$. The input constraint yields $\dot W \le \kappa(E_{\max} - W)$, hence $E_{\max} - W(t) \ge (E_{\max} - \overline{E}_0)\exp(-\kappa t) \ge 0$. The spatial estimate proved above completes the result.

This guarantee uses the full physical input port rather than the energy predicted by a two-mode learned model. A learned model can generate the requested field, but its unobserved high modes do not disappear from the work balance. The result assumes an accurate actual-port measurement and ideal command application; sensor, actuation and delay errors require additional work-error budgets. No field slew-rate guarantee is claimed. Because dissipated energy is not credited back, the work observer can restrict the request even when the actual stored energy has decreased. This is a conservative safety mechanism, not an arbitrary tracking theorem.

A bounded port-measurement error can be included without assuming an exact learned state. If $|a_p - \widehat{a}_p| \le \delta_a(t)$, replace the observer derivative and its input constraint by

$$\begin{aligned} \dot W &= \widehat{a}_p B_y + \delta_a |B_y| \\ &\le \kappa(E_{\max} - W). \end{aligned} \tag{99}$$

Actual input power is at most this observer derivative, so the same proof applies. Zero field remains feasible and the scalar

feasible set is convex. Here B_y must be the actual applied field; an uncertain command-to-field map requires a further bound on actual input power. The numerical tests below use $\delta_a = 0$ and do not experimentally validate a sensor-error bound.

7.3 Full-FE safety, violation and truncation checks

All tests use the shared 32-element consistent-mass FE beam, retain all 64 unconstrained displacement/slope degrees of freedom, and integrate the exact cosine tip potential. The coordinates are the complete 64-mode static-field basis; this is a change of coordinates, not a modal truncation. The default drag is $C = \alpha M$ with $\alpha = 0.55558524$ s^{-1}. Unless stated otherwise, trajectories start from rest at the straight equilibrium. The sinusoidal tests use the original $\Omega = 9.444949$ rad/s and the 0.25 s ramp above; the monotone tests omit the sine factor. The integration interval is 8 s. The continuous-time claims come from Propositions 9 and 10; cubic-Hermite spatial extrema are evaluated exactly at each saved time for the numerical diagnostics.

Table 7 separates numerical maxima from continuous-time certificates. The 6 mT monotone ramp satisfies an all-time displacement bound of 3.48494 mm, including zero drag. The default 6 mT sinusoid has a numerical peak of 3.09031 mm, but fails the conservative field-variation test; that peak alone is not a safety proof. Increasing its amplitude to 10 mT produces a 5.14478 mm free-motion peak and crosses the 4 mm boundary. The continuation after crossing is a diagnostic of a violated constraint, not a simulation of wall contact.

With $E_{\max} = 0.95E_c = 22.66365$ nJ and $\kappa = 5$ s^{-1}, the work governor applied to the same 10 mT request has a rigorous whole-centerline bound of 3.89872 mm and thus a minimum guaranteed clearance of 0.10128 mm. Its numerical peak is 3.14362 mm with the default drag and 3.55853 mm with zero drag. The same energy cap also implies $\sup_s |w_s| \le 0.08441$, consistent with the declared small-slope regime for the certified trajectories. Tightening both relative and absolute integration tolerances tenfold for the full-FE governed case changes the tip trajectory by at most 8.19×10^{-11} m; the refined power-balance residual is 3.86×10^{-18} J (1.74×10^{-10} relative to the work scale). The governor changes the requested field at 55.37% of saved times in this oscillatory case.

For the original 6 mT sinusoid, the maximum tip discrepancies of the 2-, 4-, and 8-mode reconstructions against the full 64-mode FE trajectory are 22.7168, 5.9661, and 1.5156 μm, respectively (0.73510%, 0.19306%, and 0.04904% of the full-FE peak). These are measured truncation errors for this input, not rigorous bounds for all neglected modes or all trajectories. The full-continuum energy certificate does not substitute any of them for an unknown error constant. An independent COMSOL HermitianBeam integration of the same 32-element consistent-mass model gives a default-sinusoid peak of 3.09032 mm; its largest tip difference from the present integration is 0.11835 μm. Tightening the COMSOL relative tolerance to 10^{-8} reduces the largest tip difference to 0.02160 μm. Independent COMSOL runs also reproduce the 6 mT monotone-ramp safe case and 10 mT sinusoidal violation: their peaks are 0.924046 and 5.144787 mm, with maximum cross-solver tip discrepancies of 0.00663 and 0.03489 μm, respectively. This cross-check concerns the same mathematical model and does not calibrate material or actuation parameters against hardware.

The closed-loop governor was also implemented inside COMSOL, coupling a global ODE for $W/E_{\max}$ to the nonlinear magnetic end moment. This run used the generalized-α time integrator with $\rho_\infty = 0$, relative tolerance 10^{-5}, and a maximum internal step of 0.002 s; the 2001 saved outputs are spaced by 0.004 s. Its full-FE centerline peak is 3.143643 mm, leaving a sampled net clearance of 0.856357 mm. The maximum tip discrepancy from the refined Python trajectory is 0.582073 μm, or 0.018516% of its peak. The applied field remains below 9.816986 mT. The COMSOL-integrated work observer peaks at 22.253032 nJ, or $0.9818823E_{\max}$; $E \le W \le E_{\max}$ and the magnetic-power projection hold at every saved time. Its maximum work-observer discrepancy from Python is 2.263×10^{-12} J.

For an independent energy ledger, reconstruct E from all displacement and velocity DOFs and compute

$$D_h(t) = \int_0^t \alpha \dot{q}^{\mathsf{T}} M \dot{q} \, \mathrm{d}\tau,$$
$$R_E(t) = E(t) - E(0) - [W(t) - W(0)] + D_h(t).$$

Simpson quadrature on the saved outputs gives $\max_t |R_E(t)| = 5.796 \times 10^{-12}$ J after rounding up, or 2.605×10^{-4} of the work scale. The Simpson–trapezoid difference for that dissipation integral is 2.424×10^{-13} J. This COMSOL run uses high-frequency numerical damping and a looser tolerance than the refined Python integration; its ledger residual includes time-integration and output-quadrature errors. The separately integrated COMSOL

Table 7. Safety tests on the same full 32-element, 64-DOF FE beam. Numerical maxima are cubic-Hermite spatial extrema at 2001 saved times over 8 s. Bounds are continuous-time sufficient bounds from the stated propositions. "No" means the sufficient test does not certify the case. All fields use a 0.25 s ramp; sine frequency is 9.444949 rad/s. Except the stated $E_c/4$ case, initial energy is zero.

Applied protocol	Peak (mm)	Bound (mm)	Certified
6 mT ramp/hold	0.92405	3.48494	Yes
3 mT ramp, $E(0) = E_c/4$	2.06222	3.05276	Yes
0.15 mT sine	0.07731	3.37863	Yes
6 mT sine	3.09031	135.145	No
10 mT sine	5.14478	225.242	No
Governed 10 mT sine	3.14362	3.89872	Yes
Governed 10 mT, zero drag	3.55853	3.89872	Yes
6 mT ramp, zero drag	0.95405	3.48494	Yes

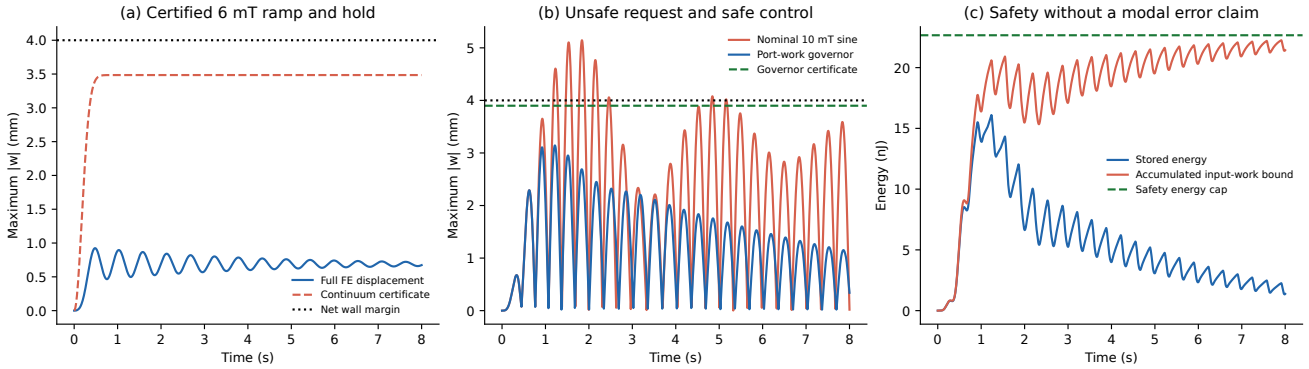


Figure 7. Whole-beam safety for the shared full FE plant, with zero initial energy and default drag. (a) A 6 mT monotone ramp and its all-time continuum energy certificate. (b) The 10 mT ramped sinusoidal request at 9.444949 rad/s violates the 4 mm net corridor margin; the actual port-work governor preserves it with an analytically guaranteed bound. (c) Stored energy and accumulated input-work bound for the governed case. The wall half-width is 4.6 mm and rod radius 0.6 mm. These are declared geometric constraints on a simulated beam, not patient-specific noncollision evidence or a KAM test.

work state is used for the observer comparison, rather than replacing it by a coarse-output input power integral. These numerical checks corroborate the implementation; the continuous-time clearance statement remains the analytic invariant in Proposition 10.

7.4 The learned CHNN generates a feasible magnetic tracking request

To connect identification to an actual input decision, the fitted GC model with seed 17 and its identified dissipation coefficient generate a nominal magnetic field for the full FE plant. Let $z = (X_1, X_2, Y_1, Y_2)$ denote the observed projection onto the first two static-field modes, $\ell^{\mathsf{T}} X$ its reconstructed tip displacement, and h_θ the actual fitted energy, including all learned mixed and momentum-dependent quartic coefficients. Define, in the dimensionless time $\tau = \omega_0 t$,

$$y = \ell^{\mathsf{T}} X, \qquad y' = \ell^{\mathsf{T}} h_{\theta,Y}, \tag{100}$$

$$f_0 = \ell^{\mathsf{T}} h_{\theta,YX} h_{\theta,Y} - \ell^{\mathsf{T}} h_{\theta,YY}(h_{\theta,X} + D_\theta h_{\theta,Y}), \tag{101}$$

$$g_0 = \ell^{\mathsf{T}} h_{\theta,YY} d \cos(d^{\mathsf{T}} X), \tag{102}$$

$$D_\theta = \widehat{\gamma}\ \mathrm{diag}(\nu)^{-1}. \tag{103}$$

Where $g_0 \neq 0$, the nominal request is

$$a_r = r'' - k_p(y - r) - k_d(y' - r'), \tag{104}$$

$$u_* = (a_r - f_0)/g_0, \tag{105}$$

$$B_{\mathrm{req}} = B_0\, \mathrm{clip}_{[-b,b]}(u_*), \tag{106}$$

$$b = B_{\max}/B_0. \tag{107}$$

The primes on r denote derivatives in τ. We use $k_p = 16/\omega_0^2$, $k_d = 8/\omega_0$, and $\widehat{\gamma} = 0.0486617964$. Both Hessian blocks are evaluated from the fitted model; they are not replaced by their quadratic-prior values. The plant receives only the

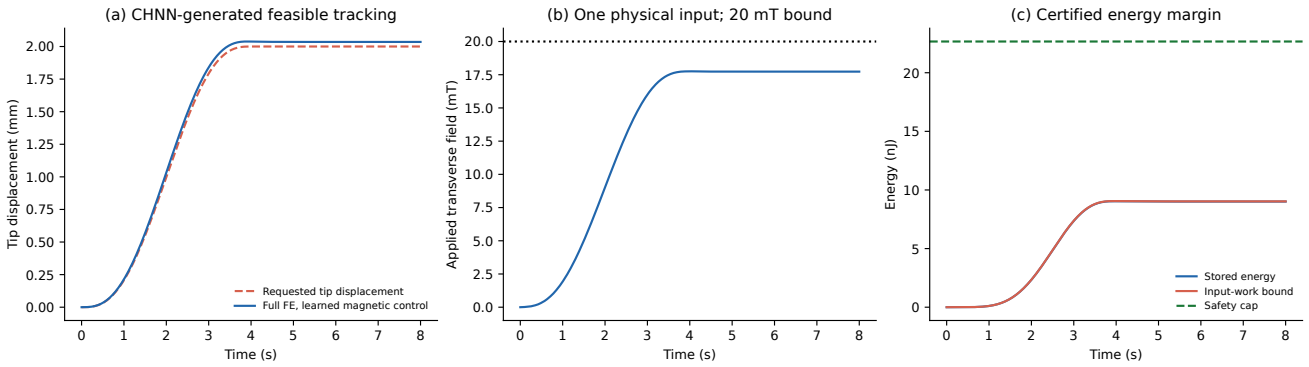


Figure 8. The fitted GC17 CHNN and identified drag generate a single bounded transverse magnetic input for the same full 32-element FE beam. (a) A feasible 2 mm tip request over 4 s and the actual full-FE tip motion. (b) The applied field remains below the declared 20 mT limit. (c) Stored energy and the actual-port work observer stay below the safety cap. The controller uses the exact first-two-mode projection available in this simulation; the safety observer uses full-model tip power. No modal actuation authority or unmeasured-state observer is assumed. The governor is inactive for this feasible tracking request; its active effect is tested separately in Figure 7.

scalar B_y through its actual tip moment, after the work-governor projection. It receives no independently assigned modal forces.

The requested SI tip trajectory is $y_{\mathrm{ref}}(t) = 2\,\mathrm{mm}\,(10s^3 - 15s^4 + 6s^5)$ for $s = t/(4\,\mathrm{s}) \in [0,1]$, followed by a hold through 8 s, and $r(\tau) = y_{\mathrm{ref}}(\tau/\omega_0)$. The field limit is $B_{\max} = 20$ mT for this separate tracking protocol. This limit is justified by the actual static input authority: a 2 mm equilibrium requires $\vartheta = 2y_{\mathrm{tip}}/L = 0.05$ and

$$B_y^{\mathrm{eq}} = \frac{EI\vartheta/(\mu L) + B_0 \sin\vartheta}{\cos\vartheta} \simeq 17.43\ \mathrm{mT}. \qquad (108)$$

The original 10 mT sinusoidal test and its safety ablation retain their original amplitudes; their actuator limit is not silently used to claim slow 2 mm tracking.

On the full 64-DOF plant, the actual tip has an RMS tracking error of 33.734 μm and a maximum error of 47.069 μm. The final tip position is 2.03515 mm and the largest centerline displacement is 2.03838 mm. The applied field peaks at 17.75695 mT; saturation and governor intervention occur at 0% of saved times. Thus the feasible learned request passes through the filter unchanged, and the separate unfiltered run is identical. The input-work bound peaks at 9.05293 nJ, below the 22.66365 nJ cap, and the same continuous-time safety invariant applies. The minimum absolute input gain along this run is 0.0138678 m, so the division in (107) does not approach a singularity. The physical power-balance residual is 3.53×10^{-20} J. This is a numerical tracking demonstration with the two-mode projection observed from the full state; an observer for incomplete sensing and a general tracking theorem are not claimed.

An additional independent run uses the frozen CHNN identified from COMSOL displacement and velocity exports by integral fitting, with cosine multiplier 1.0223248822. Its complete gradient and Hessian, including learned momentum quartics, generate the same 2 mm request on the same full-FE Python plant. For this conservative fit, a separate PSD drag identification on the existing Python-generated damped beam trajectories gives $\widehat{\gamma} = 0.0486617702$; this dissipative training source is distinct from the COMSOL conservative data. The tip RMSE is 33.726 μm, its maximum error is 47.065 μm, and the largest centerline displacement is 2.038365 mm. The applied field peaks at 17.756854 mT and the work observer at 9.052842 nJ. Saturation and governor intervention are again zero at the saved times, and the physical power-balance residual is 3.53×10^{-20} J. The saved fit and damping-file hashes bind this run to its actual readouts; the original GC17 results are preserved. The full-state projection and actual-port measurement assumptions are unchanged, and the same continuum clearance guarantee applies.

Finally, the single uniform-field dipole supplies one boundary moment. In static equilibrium with no distributed transverse loading, $w_{ssss} = 0$ and the free-tip shear condition imply $w(s) = \vartheta s^2/(2L)$. This one-parameter family cannot realize an arbitrary S-shaped centerline. The governor preserves a corridor while filtering a feasible magnetic request; richer prescribed paths require a demonstrated actuation map, such as distributed magnetic moments or additional independently controlled loads. No such authority is inferred from the scalar input used here.

8 Discussion

The mechanical, learned and operational models now share the same clamped–free beam, static magnetic field and dimensional map. The independent matrix comparison verifies the finite-element assembly; the COMSOL trajectory comparison tests a separate time solver with the full nonlinear magnetic moment. The learned two-mode model introduces two distinct errors: identification error within the reduced hypothesis class and truncation error relative to the full discretized beam. Reporting both prevents an accurate fit to a reduced target from being mistaken for an equally accurate approximation of every physical observable. In particular, tip-angle errors converge more slowly with modal order than tip-displacement errors.

Complex and real implementations are compared at matched function class and parameter count. Equivalent real and complex feature realizations should return equivalent fitted energies. Any benefit attributed to a mechanical prior is assessed separately from this representation choice. The conservative and dissipative readouts also have separate roles: the former determines the Hamiltonian vector field and its twist, while the latter guarantees nonpositive loss power. A small dissipative force cannot simply be relabelled as a Hamiltonian perturbation to invoke classical KAM theory.

The invariant-torus and clearance statements answer different questions. KAM persistence concerns quasiperiodic conservative motion on invariant tori with the stated nonresonance and regularity assumptions. It does not imply that every initial condition is quasiperiodic or that resonance necessarily causes chaos. The continuum energy estimate instead bounds all configurations that obey its energy budget, including driven dissipative trajectories. The input governor uses the actual magnetic power port to preserve that budget. Consequently its guarantee does not rely on treating the empirical two-mode error as a rigorous full-beam error bound.

The dimensional parameters specify an illustrative slender beam rather than a calibrated catheter. Material uncertainty, a nonuniform magnetic map, flowing blood, moving walls and large rotations require their corresponding terms in the rod reduction and power balance. The present planar validation does not test these uninstantiated operating conditions. The general force formulation identifies how they enter, and the safety theorem shows which additional work or modelling-error bounds must be supplied. No claim of clinical safety, arbitrary three-dimensional path controllability or in-orbit experimental validation follows from the reported simulations.

9 Conclusion

The clamped–free magnetic model provides a common basis for complex Hamiltonian learning, nonlinear invariant-motion analysis and dimensional actuation constraints. An explicit energy–dissipation decoder and loss-to-norm estimates connect fitted coefficients to stability hypotheses. Computer-assisted a posteriori proofs certify millimetre-scale invariant tori of the complete analytic target and actual learned Hamiltonians, including one function that also passes the nonzero-loss twist bound. Independent COMSOL and Python calculations verify the same finite-element model and quantify reduction error. Nondegenerate certified tori support positive-measure families in sufficiently small surrounding tubes. The unforced physical finite-element model also has a global asymptotic stability guarantee under positive damping. A continuum clearance inequality and a realizable magnetic input governor complement these results with an all-time energy-based operating guarantee. The resulting manuscript supplies mathematical arguments, actual numerical computations and reproducible model files for this declared mechanical setting.

Data and code availability

The accompanying reproducibility package contains native COMSOL models, MATLAB construction sources using the COMSOL API, exported matrices and trajectories, Python data-generation and learning code, saved observations and fitted coefficients, verification records and figure sources. Training data and independent COMSOL validation data are labelled by their actual generation process. Numerical values reported as simulations were produced by the accompanying computations.

Declarations

Author identities, affiliations, funding, competing interests and author contributions must be completed by the submitting authors. This work reports mathematical analysis and computer-generated mechanical data. No human, animal or in-orbit experiment was performed.

A Integrator structure and reproducibility checks

For a real smooth Hamiltonian, implicit midpoint is

$$x_{k+1} = x_k + hJ\nabla H\big((x_k + x_{k+1})/2\big). \tag{109}$$

Let $A = JH''(\bar{x})$. Differentiation gives $D\Phi_h = (I - hA/2)^{-1}(I + hA/2)$. Since $A^{\mathsf{T}}J + JA = 0$, this Cayley transform satisfies $D\Phi_h^{\mathsf{T}} J D\Phi_h = J$ whenever the implicit equation defines a differentiable local map. The proof assumes the nonlinear step is solved exactly; residual tolerances add a numerical defect. Explicit RK4 is not symplectic in general. For the separable physical Hamiltonian, the kick–drift–kick Störmer–Verlet map is also symplectic, because each substep is the exact flow of the kinetic or potential Hamiltonian. A general learned momentum-dependent quartic no longer admits that separable split, so the same formula cannot be reused without justification.

Energy error and symplecticity are distinct diagnostics. Finite-time energy plots are evaluated against the specified reference Hamiltonian and accompanied by a step-size check. A small integration error does not certify a Diophantine inequality, nor does a positive finite-time Lyapunov estimate alone establish asymptotic chaos. These facts determine the labels used in the numerical figures.

B The fourth-order central generating-function scheme

Let $f = J\nabla H$, $A = H''$ and define a central generating function

$$S_h(x) = hH(x) + \frac{h^3}{24} f(x)^{\mathsf{T}} A(x) f(x). \tag{110}$$

The map is specified implicitly by

$$x_+ - x_- = J\nabla S_h\big((x_- + x_+)/2\big). \tag{111}$$

Canonical maps generated in central variables are a standard geometric integration construction (Feng et al. 1989; Hairer et al. 2006). Here every derivative is computed from the same learned scalar Hamiltonian. The identity

$$J\nabla(f^{\mathsf{T}} A f) = f''[f,f] - 2(f')^2 f \tag{112}$$

follows by differentiating both occurrences of f, using $f' = JA$ and $J^{\mathsf{T}} = -J$. It gives the displacement implemented in the reproducibility code:

$$g_h = hf + \frac{h^3}{24}\big(f''[f,f] - 2(f')^2 f\big). \tag{113}$$

The Hessian S_h'' is symmetric, so the midpoint Cayley-transform argument in Appendix A proves exact symplecticity of the locally defined map. Since $S_{-h} = -S_h$, interchanging its endpoints proves reversibility.

To verify order, expand an exact orbit about its temporal midpoint x_0. Its endpoint midpoint m and endpoint difference satisfy

$$m = x_0 + \tfrac{1}{8}h^2 f'(x_0) f(x_0) + O(h^4), \tag{114}$$

$$x_+ - x_- = hf(x_0) + \tfrac{1}{24}h^3\big(f''[f,f] + (f')^2 f\big)(x_0) + O(h^5). \tag{115}$$

Substituting $x_0 = m - \frac{1}{8}h^2 f'(m) f(m) + O(h^4)$ gives $x_+ - x_- = g_h(m) + O(h^5)$, proving fourth order under the required smoothness and local solvability assumptions.

B.1 An integrator check on the actual learned magnetic energy

We implement (111) for the frozen primary GC17 Hamiltonian, retaining its complete cosine and all mixed and momentum-dependent quartics. Polynomial and trigonometric derivatives through order four are evaluated analytically from the saved energy; they supply both the central displacement and its Newton Jacobian. This test closes all forcing and damping ports. Starting at $(X_1, X_2, Y_1, Y_2) = (0.1, 0.06, 0, 0.02)$, we integrate to $\tau = 10$ and compare against DOP853 with relative and absolute tolerances 3×10^{-14} and 3×10^{-16}. Tightening the reference tolerances from 2×10^{-13} and 2×10^{-15} changes the sampled reference by at most 8.32×10^{-14} per state component. The endpoint errors in Table 8 approach fourth order for Central4 and second order for implicit midpoint.

The largest implicit-step residual in the infinity norm is 2.00×10^{-15}, with at most three Newton evaluations per step. For each tested step size, differentiating the converged implicit equation gives its one-step tangent map $D\Phi_h$. The largest measured defect $\|D\Phi_h^{\mathsf{T}} J D\Phi_h - J\|_\infty$ is 3.34×10^{-16} for Central4 and 2.24×10^{-16} for midpoint. An independent fourth-order central-difference evaluation of the solved step map agrees with the implicit tangent to 1.39×10^{-12}; its largest symplectic defect is 2.75×10^{-12}. Thus both methods preserve the symplectic structure to the measured implicit-solve and arithmetic accuracy, while their convergence orders differ. These are numerical checks of the stated algorithms, not a new KAM certificate or a separable Verlet approximation to a momentum-dependent learned Hamiltonian.

C Complex-gradient descent with a strict descent margin

Theorem 3 of Zhang et al. (2022) motivates complex-gradient optimization, but a stationary-point conclusion

Table 8. Conservative integration of the same saved GC17 energy. Errors are Euclidean endpoint errors in the balanced dimensionless state; orders compare successive step halvings.

Step h	Midpoint error	Order	Central4 error	Order
0.08	7.1351×10^{-2}	—	1.8346×10^{-3}	—
0.04	1.9383×10^{-2}	1.8802	1.1784×10^{-4}	3.9606
0.02	4.8966×10^{-3}	1.9849	7.4149×10^{-6}	3.9902
0.01	1.2265×10^{-3}	1.9972	4.6422×10^{-7}	3.9976

requires a strict descent margin. The following convention-explicit statement also applies to a smooth CHNN regression objective.

Proposition 11. Complex descent. *Let $\mathcal{L} : \mathbb{C}^P \simeq \mathbb{R}^{2P} \to \mathbb{R}$ be bounded below and have an L-Lipschitz real gradient on every segment traversed by the iterates. For $w = a + ib$, define $G(w) = \nabla_a \mathcal{L} + i\nabla_b \mathcal{L} = 2\partial\mathcal{L}/\partial\bar{w}$. Choose a constant $\rho \in \mathbb{C}$ with $\kappa = \operatorname{Re}\rho - (L/2)|\rho|^2 > 0$, and update $w_{t+1} = w_t - \rho G(w_t)$. Then*

$$\mathcal{L}(w_{t+1}) \le \mathcal{L}(w_t) - \kappa\|G(w_t)\|_2^2, \qquad \lim_{t\to\infty} \|G(w_t)\|_2 = 0,$$

and

$$\min_{0\le t<T} \|G(w_t)\|_2^2 \le \frac{\mathcal{L}(w_0) - \inf \mathcal{L}}{\kappa T}.$$

Proof. Under realification, multiplication by $\rho = u + iv$ is $uI + vJ_c$, with $J_c^\top = -J_c$ and $J_c^\top J_c = I$. Hence the update has inner product $-u\|G\|^2$ with the real gradient and squared norm $|\rho|^2\|G\|^2$. The real descent lemma gives the first inequality. Summing it and using the lower bound for $\mathcal{L}$ gives $\kappa \sum_{t\ge 0} \|G(w_t)\|^2 \le \mathcal{L}(w_0) - \inf \mathcal{L}$. Square summability proves convergence of the gradient norm and the minimum bound. The result asserts convergence to stationarity in gradient norm, not convergence to a global minimum or to a particular parameter vector.

For a real step this condition is $0 < \rho < 2/L$ under the stated convention. The case $\rho = 0$, allowed by a non-strict printed step-size interval, cannot imply gradient convergence. Nonsmooth complex activations also need separate treatment; a smooth polynomial CHNN satisfies the differentiability requirements, with a Lipschitz gradient bound on any bounded iterate region.

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